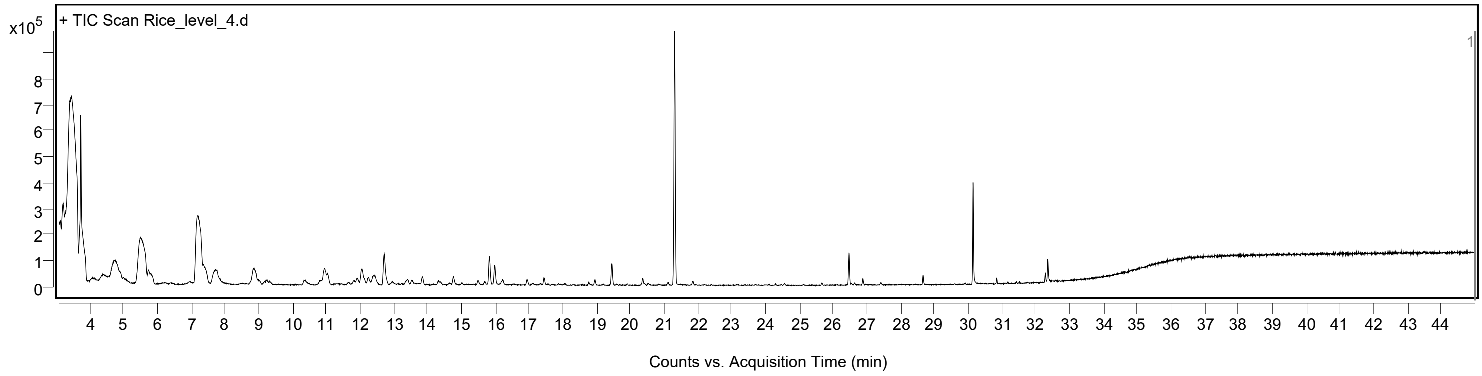
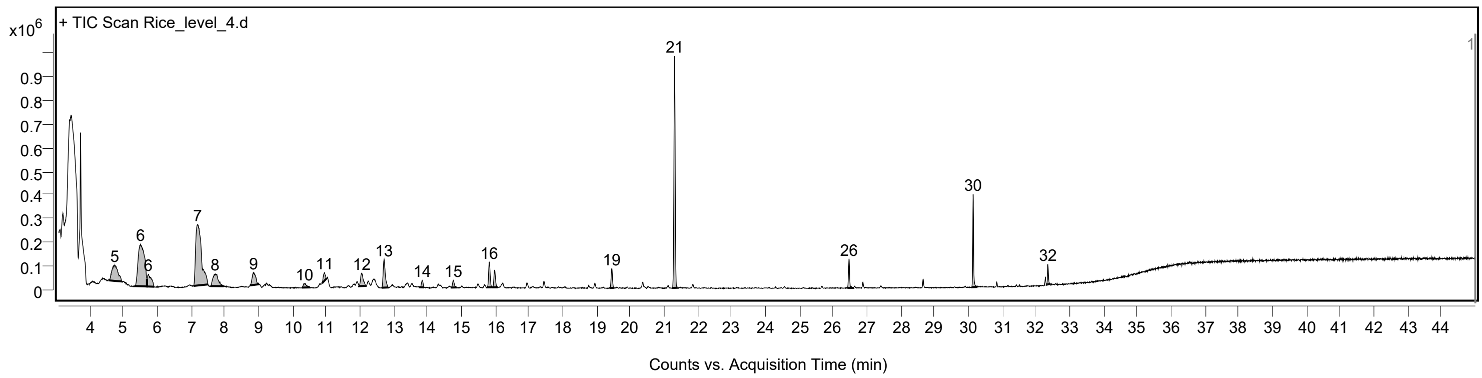
Analysis Report

Sample Information

Sample Name	Rice_level_4	**Data File Path**	D:\MassHunter\GCMS\1\data\20260721 Rice leaf\20260722 RUN SEQUENCE\Rice_level_4.D
Sample ID		**Acq Time (Local)**	22/7/2569 19:38:35 (UTC+07:00)
Instrument	GCMS_HSS	**Acq Method Path**	D:\MassHunter\GCMS\1\methods\RICE_LEAF_HS.M
MS Type	Q	**Acq SW Version**	MassHunter GC/MS Acquisition 10.2.530.3 21-Nov-2023 Copyright © 1989-2023 Agilent Technologies, Inc.
Inj Vol (ul)	1	**IRM Status**	
Sample Position	28	**DA Method Path**	D:\MassHunter\GCMS\1\data\20260721 Rice leaf\20260722 RUN SEQUENCE\Rice_level_4.D\Results\Qual\Version4\default.m
Plate Position		**Target Source Path**	
Acq Operator		**Result Summary**	

Sample Chromatograms



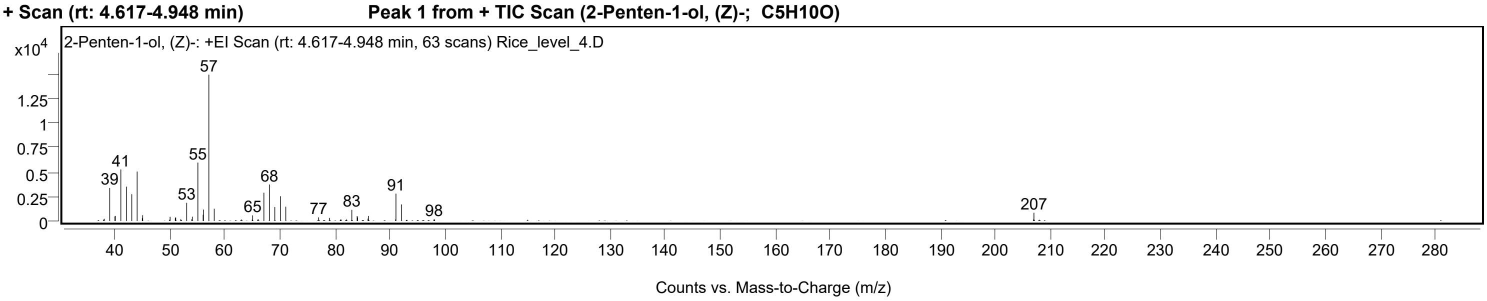


Chromatogram Peaks

Peak	Start	RT	End	Height	Area	Area %	SNR
1	4.595	4.756	4.959	65756	823961	26.84	
2	5.352	5.521	5.708	176277	2333839	76.03	
3	5.708	5.745	5.922	50525	415513	13.54	
4	7.085	7.206	7.515	256602	3069702	100.00	
5	7.588	7.724	7.976	51418	567634	18.49	
6	8.755	8.864	9.019	51022	381547	12.43	
7	10.295	10.367	10.522	16280	114447	3.73	
8	10.886	10.960	11.012	32315	140023	4.56	
9	11.982	12.062	12.174	54473	295743	9.63	
10	12.656	12.720	12.843	117985	517996	16.87	
11	13.774	13.849	13.906	29038	100468	3.27	
12	14.720	14.769	14.865	29754	105408	3.43	
13	15.763	15.833	15.903	106080	336508	10.96	
14	15.908	15.993	16.082	72752	238211	7.76	
15	19.401	19.454	19.513	81114	227737	7.42	
16	21.241	21.315	21.412	975162	2555591	83.25	
17	26.424	26.472	26.600	121238	291729	9.50	
18	30.103	30.141	30.242	390346	682609	22.24	
19	32.318	32.355	32.408	84058	159990	5.21	

Sample Spectra

+ Scan (rt: 4.617-4.948 min) **Peak 1 from + TIC Scan (2-Penten-1-ol, (Z)-; C5H10O)**



Spectrum Peaks

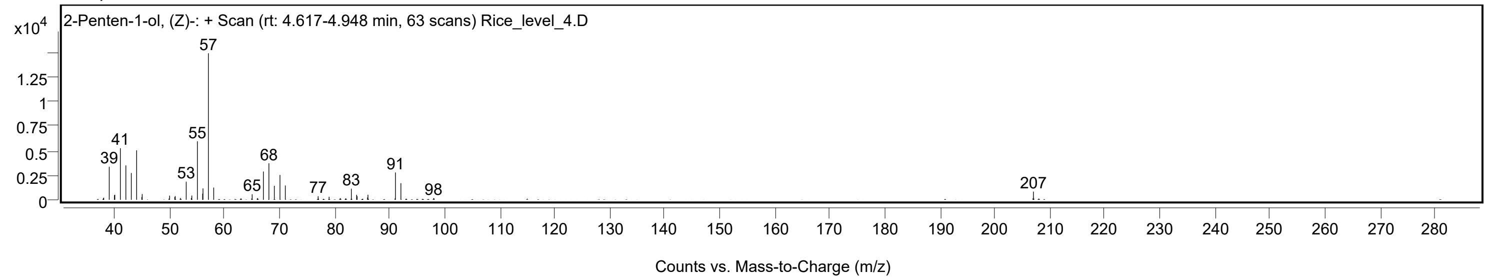
m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
38.0200		221	1.48					
39.0200		3365	22.59					
39.9600		473	3.17					
40.0400		516	3.46					
41.0400		5245	35.21					
42.0400		3494	23.46					
43.0300		2728	18.31					
44.0000		5028	33.75					
44.9400		282	1.90					
45.0100		589	3.95					
49.9900		425	2.85					
50.9300		293	1.97					
51.0200		379	2.54					
51.9400		182	1.22					
53.0100		1846	12.39					
53.9300		174	1.17					
54.0300		434	2.91					
55.0400		5943	39.89					
55.9900		631	4.24					
56.0600		1160	7.79					
57.0400		14898	100.00					
58.0100		1246	8.36					
65.0000		571	3.83					
66.0100		172	1.15					
67.0400		2874	19.29					
68.0400		3713	24.92					
69.0200		1426	9.57					
70.0500		2522	16.93					
71.0200		1454	9.76					
76.9900		370	2.48					
78.9800		311	2.09					
81.0400		164	1.10					
82.0200		152	1.02					
83.0200		1132	7.60					
83.9800		509	3.42					
84.0700		367	2.46					
85.9600		239	1.60					
86.0400		513	3.44					
91.0400		2778	18.65					
92.0300		1689	11.34					
97.9600		169	1.13					
206.9900		834	5.60					

Spectrum Identification Table

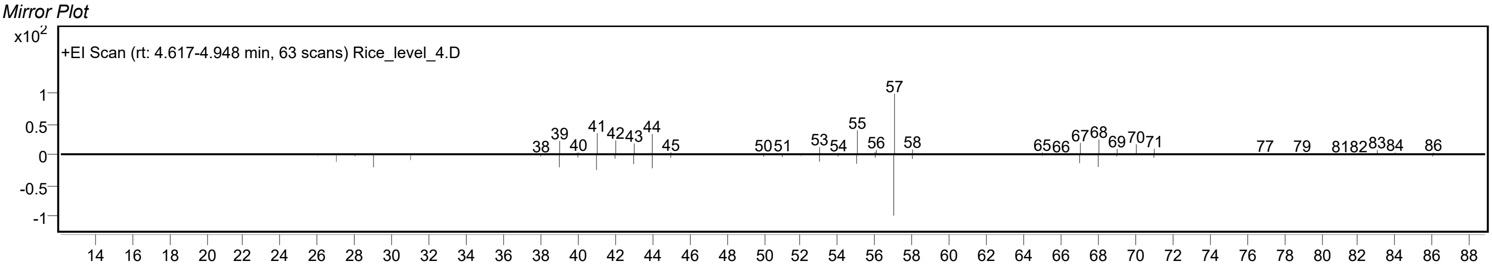
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	2-Penten-1-ol, (Z)-	C5H10O				1576-95-0	59.62	59.62			NIST23.L
No	LibSearch	2-Penten-1-ol, (Z)-	C5H10O				1576-95-0	58.80	58.80			NIST23.L
No	LibSearch	2-Penten-1-ol, (Z)-	C5H10O				1576-95-0	57.96	57.96			NIST23.L
No	LibSearch	2-Methylidenebutan-1-ol	C5H10O				4435-54-5	57.04	57.04			NIST23.L
No	LibSearch	2-Penten-1-ol, (E)-	C5H10O				1576-96-1	57.01	57.01			NIST23.L
No	LibSearch	2-Penten-1-ol, (Z)-	C5H10O				1576-95-0	56.77	56.77			NIST23.L
No	LibSearch	Cyclopentanol	C5H10O				96-41-3	55.29	55.29			NIST23.L
No	LibSearch	(3-Ethyloxetan-3-yl)methanol	C6H12O2				3047-32-3	54.98	54.98			NIST23.L
No	LibSearch	2-Penten-1-ol, (E)-	C5H10O				1576-96-1	54.55	54.55			NIST23.L
No	LibSearch	2-Hepten-1-ol, (E)-	C7H14O				33467-76-4	53.77	53.77			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	2-Penten-1-ol, (Z)-	C5H10O		1576-95-0	12.700		59.62	59.62			NIST23.L

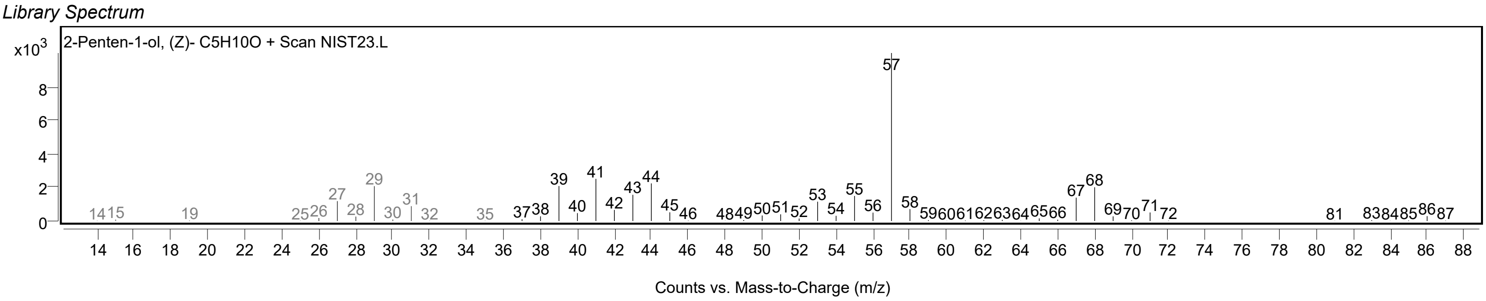
Observed Spectrum



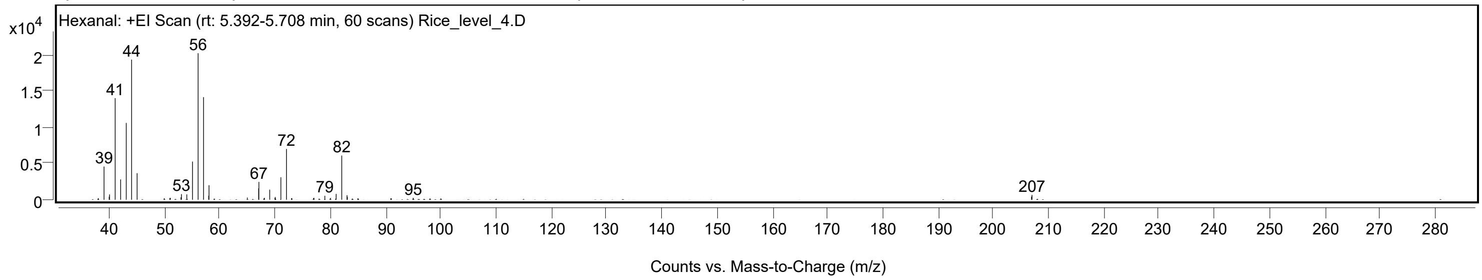
Mirror Plot



Library Spectrum



+ Scan (rt: 5.392-5.708 min) **Peak 2 from + TIC Scan (Hexanal; C6H12O)**



Spectrum Peaks

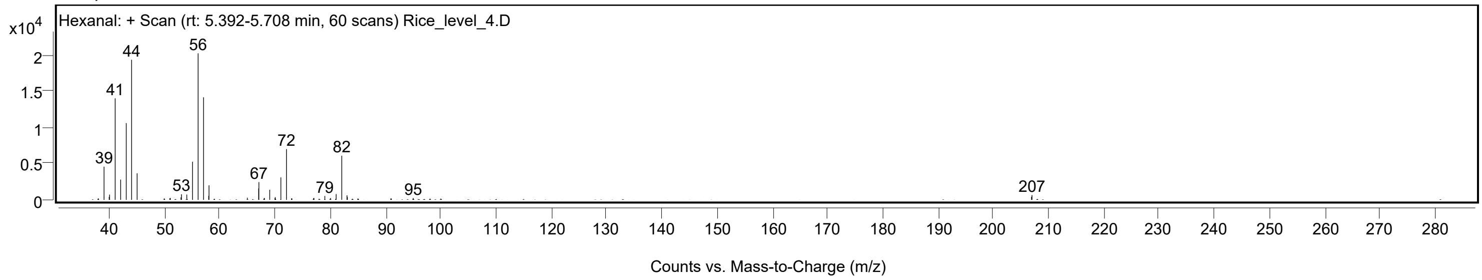
m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
37.9900		219	1.08					
39.0300		4576	22.64					
39.9600		427	2.11					
40.0300		730	3.61					
41.0500		14001	69.28					
42.0400		2798	13.85					
43.0500		10587	52.39					
44.0200		19321	95.61					
45.0200		3663	18.13					
50.9400		224	1.11					
51.0300		254	1.26					
52.9600		341	1.68					
53.0400		778	3.85					
54.0100		736	3.64					
55.0400		5264	26.05					
56.0600		20209	100.00					
57.0400	1	14146	70.00					
57.9700	1	554	2.74					
58.0300		2004	9.91					
64.9700		297	1.47					
67.0200		1570	7.77					
67.0600		2457	12.16					
68.0600		261	1.29					
69.0300		1392	6.89					
69.9600		248	1.23					
70.0300		378	1.87					
71.0400		3096	15.32					
72.0500	1	7002	34.65					
73.0400	1	206	1.02					
76.9300		203	1.00					
77.0300		262	1.30					
79.0000		543	2.69					
80.0300		245	1.21					
81.0500		802	3.97					
82.0700	1	6090	30.14					
83.0100		626	3.10					
83.0800	1	384	1.90					
95.0000		261	1.29					
206.9300		427	2.11					
207.0200		664	3.29					

Spectrum Identification Table

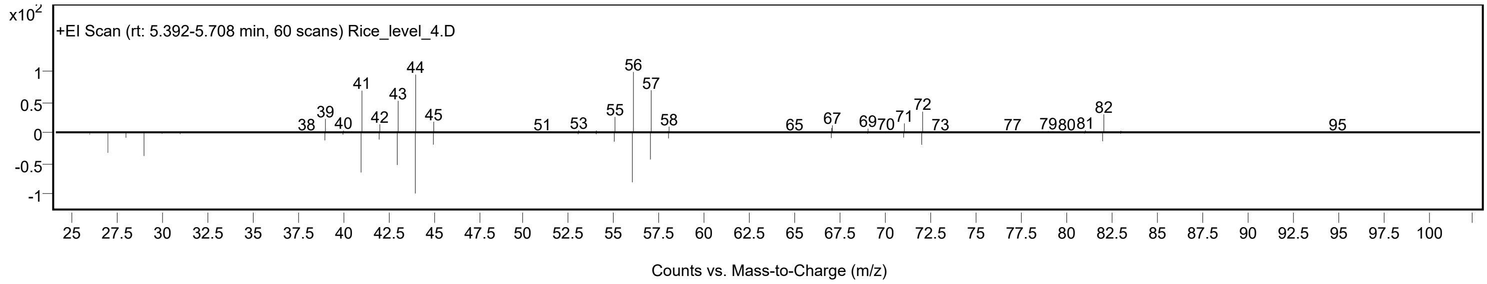
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	Hexanal	C6H12O				66-25-1	74.50	74.50			NIST23.L
No	LibSearch	Hexanal	C6H12O				66-25-1	73.26	73.26			NIST23.L
No	LibSearch	Hexanal	C6H12O				66-25-1	72.54	72.54			NIST23.L
No	LibSearch	Hexanal	C6H12O				66-25-1	72.25	72.25			NIST23.L
No	LibSearch	Hexanal	C6H12O				66-25-1	70.75	70.75			NIST23.L
No	LibSearch	Hexanal	C6H12O				66-25-1	69.09	69.09			NIST23.L
No	LibSearch	Cyclobutanol, 2-ethyl-	C6H12O				35301-43-0	68.17	68.17			NIST23.L
No	LibSearch	Pentanal, 3-methyl-	C6H12O				15877-57-3	66.86	66.86			NIST23.L
No	LibSearch	Pentanal, 3-methyl-	C6H12O				15877-57-3	66.22	66.22			NIST23.L
No	LibSearch	Pentanal, 3-methyl-	C6H12O				15877-57-3	65.99	65.99			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	Hexanal	C6H12O		66-25-1	13.400		74.50	74.50			NIST23.L

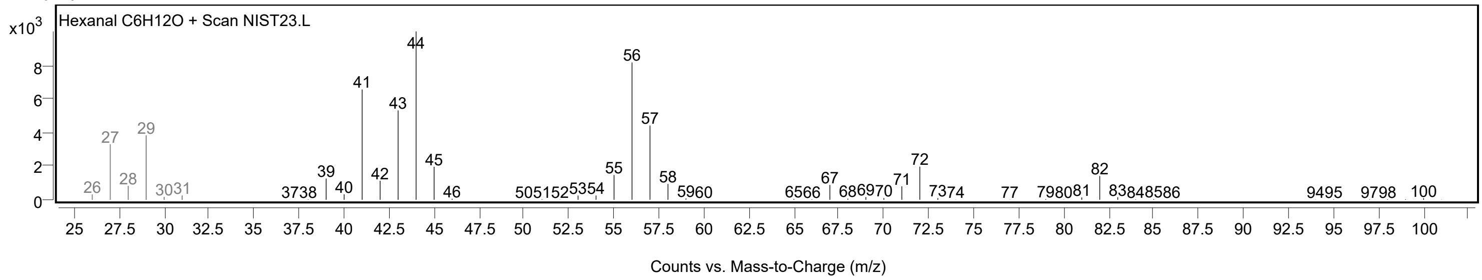
Observed Spectrum



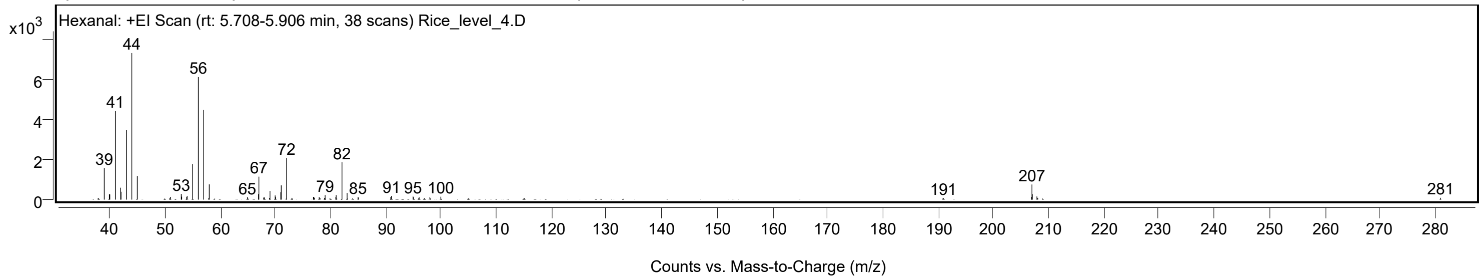
Mirror Plot



Library Spectrum



+ Scan (rt: 5.708-5.906 min) **Peak 3 from + TIC Scan (Hexanal; C6H12O)**



Spectrum Peaks

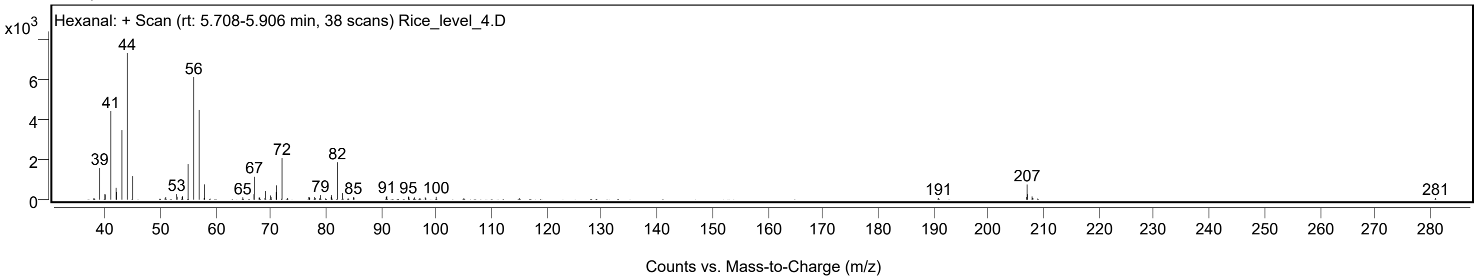
m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
39.0200		1571	21.49					
39.9400		268	3.67					
40.0300		259	3.54					
41.0300		4413	60.37					
41.9900		596	8.15					
42.0600		393	5.38					
43.0400		3462	47.36					
44.0100		7311	100.00					
45.0000		1176	16.09					
50.9900		138	1.88					
52.9600		281	3.84					
53.0500		149	2.03					
53.9400		132	1.81					
54.0300		176	2.41					
55.0300		1775	24.28					
56.0500		6113	83.62					
57.0300		4471	61.16					
58.0300		759	10.38					
64.9400		113	1.54					
66.9500		271	3.70					
67.0200	1	1146	15.68					
67.8800	1	103	1.41					
68.0100		108	1.48					
68.9300		74	1.02					
69.0300	1	434	5.94					
69.9600		208	2.85					
70.0700		125	1.71					
70.9800		368	5.04					
71.0600		710	9.71					
72.0400	1	2078	28.42					
72.9900	1	89	1.21					
76.8900		133	1.82					
76.9800		134	1.83					
77.0600		91	1.24					
77.9300		127	1.74					
78.0900		75	1.03					
78.8900		75	1.02					
79.0100		226	3.09					
80.9400		132	1.80					
81.0100		221	3.02					
82.0600		1861	25.46					
82.9900		335	4.58					
84.9600		120	1.64					
85.0800		105	1.44					
90.9000		131	1.80					
91.0300		181	2.47					
94.9200		155	2.12					
95.0300		121	1.66					
96.0400		111	1.51					
97.9700		112	1.54					
99.9600		147	2.00					
190.9500		79	1.08					
206.9000		137	1.87					
206.9900	1	755	10.33					
207.0600		269	3.68					
207.8900		159	2.18					
208.0500	1	91	1.24					
280.9800		92	1.26					

Spectrum Identification Table

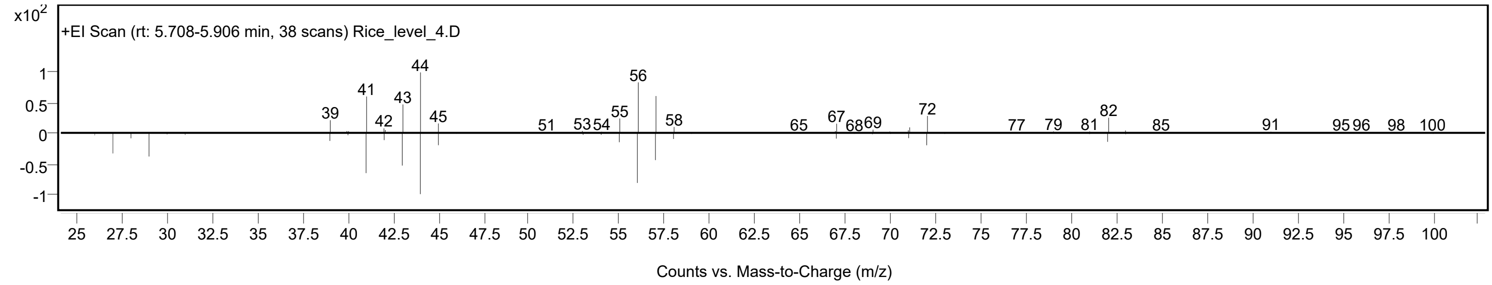
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	Hexanal	C6H12O				66-25-1	68.37	68.37			NIST23.L
No	LibSearch	Hexanal	C6H12O				66-25-1	66.38	66.38			NIST23.L
No	LibSearch	Hexanal	C6H12O				66-25-1	65.80	65.80			NIST23.L
No	LibSearch	Hexanal	C6H12O				66-25-1	65.24	65.24			NIST23.L
No	LibSearch	Hexanal	C6H12O				66-25-1	64.13	64.13			NIST23.L
No	LibSearch	Hexanal	C6H12O				66-25-1	62.43	62.43			NIST23.L
No	LibSearch	Cyclopentanol, 2-methyl-	C6H12O				24070-77-7	62.08	62.08			NIST23.L
No	LibSearch	Cyclopentanol, 2-methyl-, cis-	C6H12O				25144-05-2	60.85	60.85			NIST23.L
No	LibSearch	Pentanal, 3-methyl-	C6H12O				15877-57-3	60.82	60.82			NIST23.L
No	LibSearch	Pentanal, 3-methyl-	C6H12O				15877-57-3	60.76	60.76			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	Hexanal	C6H12O		66-25-1	13.400		68.37	68.37			NIST23.L

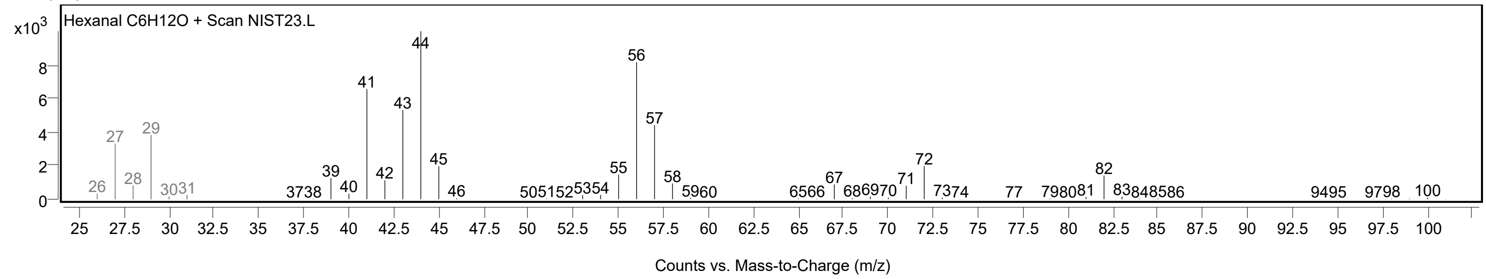
Observed Spectrum



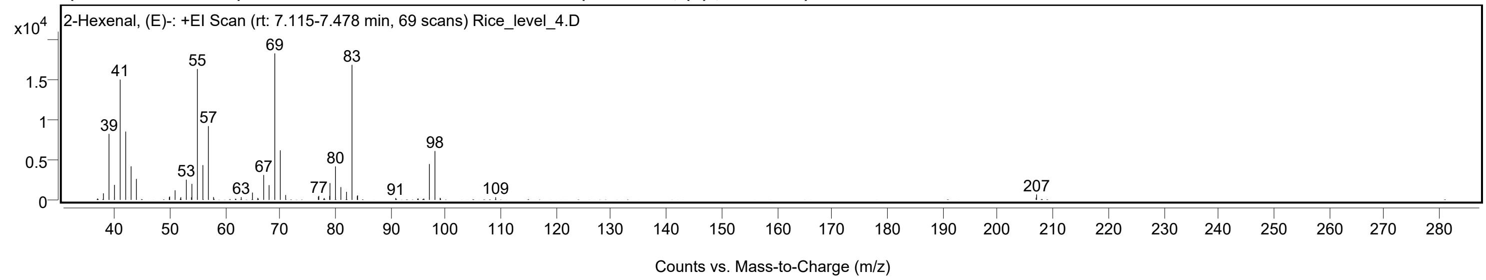
Mirror Plot



Library Spectrum



+ Scan (rt: 7.115-7.478 min) **Peak 4 from + TIC Scan (2-Hexenal, (E)-; C6H10O)**



Spectrum Peaks

m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
38.0200		826	4.52					
39.0300		8238	45.06					
40.0200		1878	10.27					
41.0500		15013	82.12					
42.0500		8529	46.65					
43.0300		4189	22.91					
43.9800		2621	14.34					
49.9500		356	1.94					
50.0200		421	2.30					
51.0000		1202	6.57					
52.0300		341	1.86					
53.0200		2525	13.81					
53.9600		330	1.81					
54.0400		2003	10.96					
55.0300		16332	89.34					
56.0300		4343	23.75					
57.0200	1	9208	50.37					
57.9600	1	352	1.92					
62.9700		345	1.89					
65.0100		916	5.01					
65.9700		247	1.35					
67.0400		3114	17.03					
68.0200		1847	10.11					
69.0500		18282	100.00					
70.0400		6183	33.82					
71.0200		612	3.35					
76.9400		414	2.26					
77.0300		501	2.74					
78.0100		229	1.25					
78.9800		485	2.65					
79.0500		2101	11.49					
80.0500		4178	22.85					
81.0200		1604	8.78					
82.0500		1010	5.52					
83.0400	1	16852	92.18					
83.9900		494	2.70					
84.0500	1	570	3.12					
90.9400		238	1.30					
97.0500		4479	24.50					
98.0600	1	6097	33.35					
98.9900	1	261	1.43					
109.0500		336	1.84					
206.9300		282	1.54					
207.0100		682	3.73					

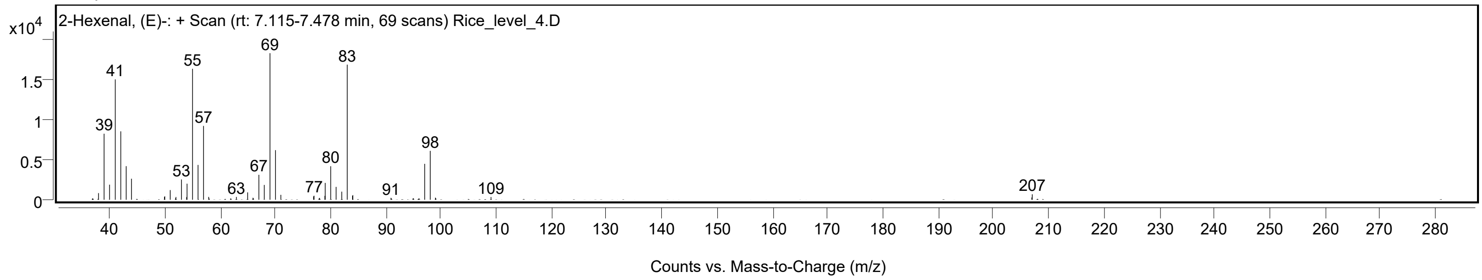
Analysis Report

Spectrum Identification Table

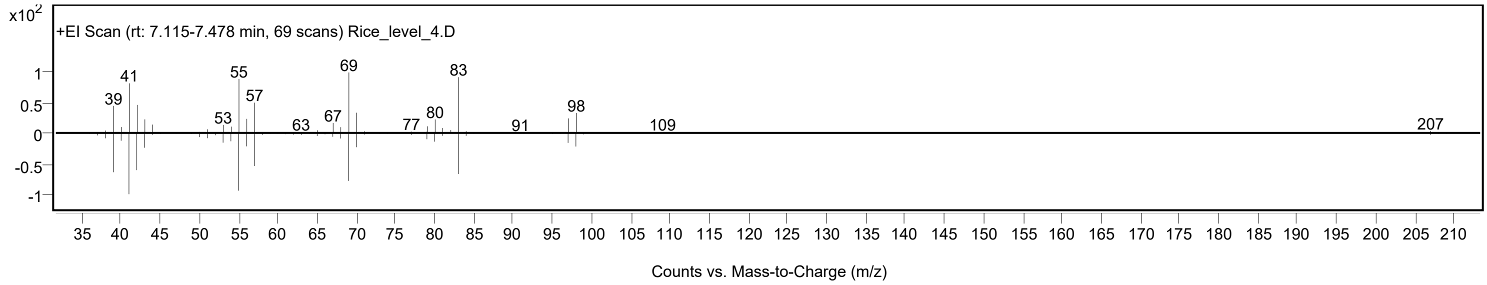
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	2-Hexenal, (E)-	C6H10O				6728-26-3	74.70	74.70			NIST23.L
No	LibSearch	2-Hexenal, (E)-	C6H10O				6728-26-3	74.62	74.62			NIST23.L
No	LibSearch	2-Hexenal, (E)-	C6H10O				6728-26-3	73.63	73.63			NIST23.L
No	LibSearch	2-Hexenal	C6H10O				505-57-7	73.39	73.39			NIST23.L
No	LibSearch	2-Hexenal, (E)-	C6H10O				6728-26-3	72.85	72.85			NIST23.L
No	LibSearch	2-Hexenal, (E)-	C6H10O				6728-26-3	72.83	72.83			NIST23.L
No	LibSearch	2-Hexenal	C6H10O				505-57-7	72.67	72.67			NIST23.L
No	LibSearch	2-Hexenal	C6H10O				505-57-7	71.76	71.76			NIST23.L
No	LibSearch	2-Hexenal, (E)-	C6H10O				6728-26-3	71.20	71.20			NIST23.L
No	LibSearch	3-Methylpenta-1,3-diene-5-ol, (E)-	C6H10O				1572-08-3	70.66	70.66			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	2-Hexenal, (E)-	C6H10O		6728-26-3	14.150		74.70	74.70			NIST23.L

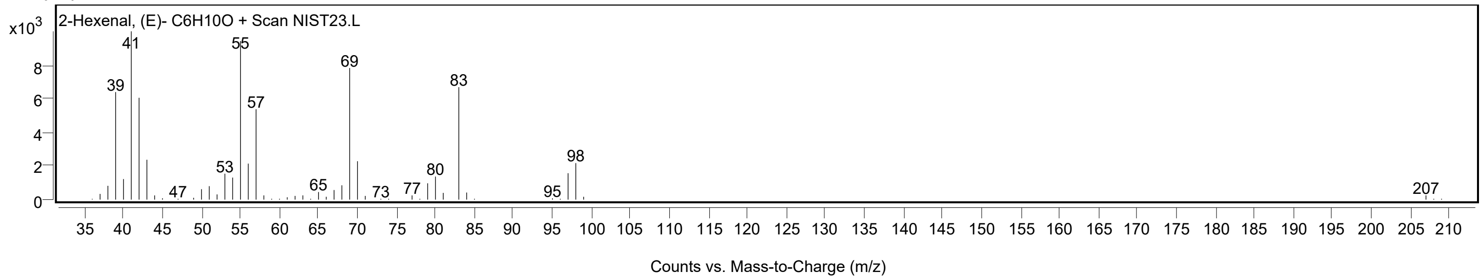
Observed Spectrum



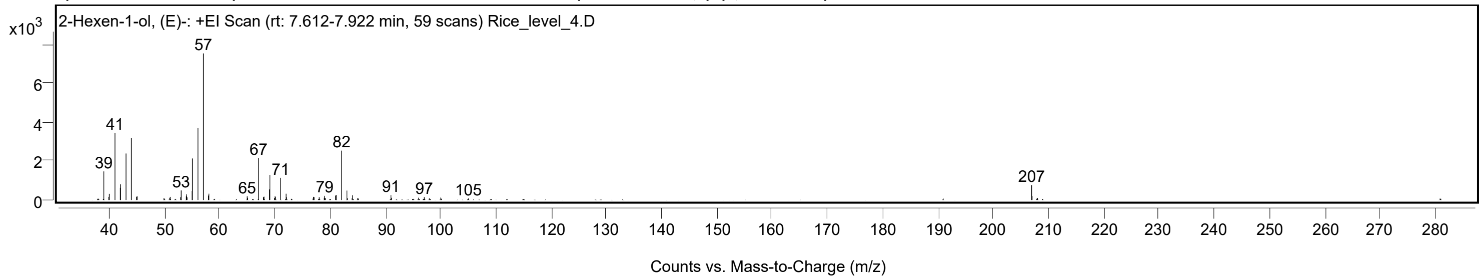
Mirror Plot



Library Spectrum



+ Scan (rt: 7.612-7.922 min) **Peak 5 from + TIC Scan (2-Hexen-1-ol, (E)-; C6H12O)**



Spectrum Peaks

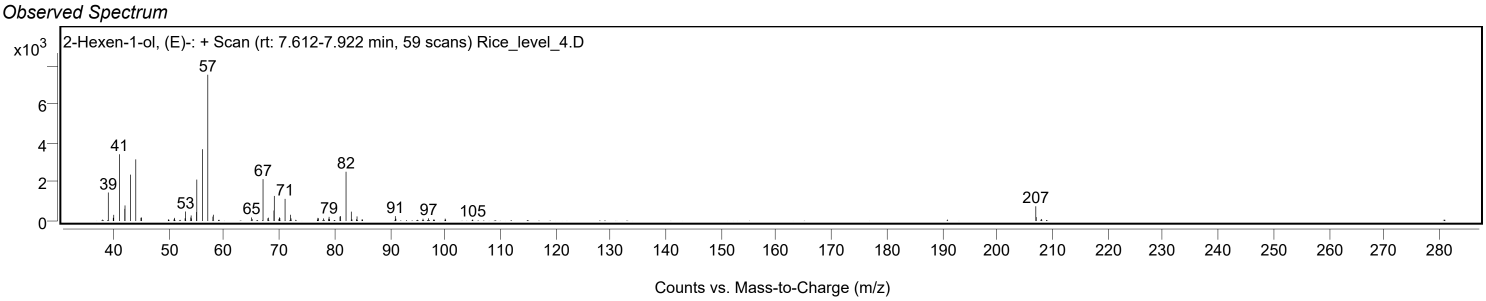
m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
39.0100		1470	19.51					
39.9100		164	2.17					
40.0000		319	4.23					
41.0300		3439	45.64					
41.9900		629	8.35					
42.0400		803	10.66					
43.0300		2390	31.72					
43.9900	1	3175	42.14					
44.9200	1	168	2.23					
45.0400		183	2.43					
49.8800		82	1.09					
50.9300		97	1.28					
51.0200		162	2.15					
52.9100		138	1.84					
53.0100		491	6.51					
53.9400		183	2.43					
54.0100		287	3.81					
54.0800		133	1.76					
54.9900		463	6.15					
55.0400		2135	28.33					
56.0500		3698	49.08					
57.0300	1	7534	100.00					
57.9500		229	3.03					
58.0300	1	324	4.30					
64.9600		185	2.45					
65.0600		101	1.35					
67.0300	1	2158	28.64					
67.9000		138	1.84					
68.0100	1	171	2.27					
69.0000	1	542	7.19					
69.0700		1293	17.16					
69.9100		110	1.46					
70.0000		189	2.51					
70.0800		153	2.03					
71.0200		1141	15.14					
72.0000		322	4.28					
72.1000		136	1.81					
76.9600		164	2.17					
77.0700		138	1.84					
77.9900		135	1.79					
78.9000		116	1.53					
78.9900		216	2.86					
79.0900		90	1.20					
80.9600		227	3.02					
81.0700		253	3.35					
82.0600		2538	33.69					
83.0100		484	6.42					
83.1400		84	1.12					
83.9300		87	1.15					
84.0500		237	3.15					
84.9600		86	1.14					
90.9800		253	3.36					
91.0900		115	1.52					
96.0000		118	1.57					
97.0100		139	1.85					
98.0000		82	1.09					
100.0100		123	1.64					
105.0000		77	1.03					
206.9600	1	759	10.07					
207.0400		202	2.68					
208.0200	1	106	1.41					

Spectrum Identification Table

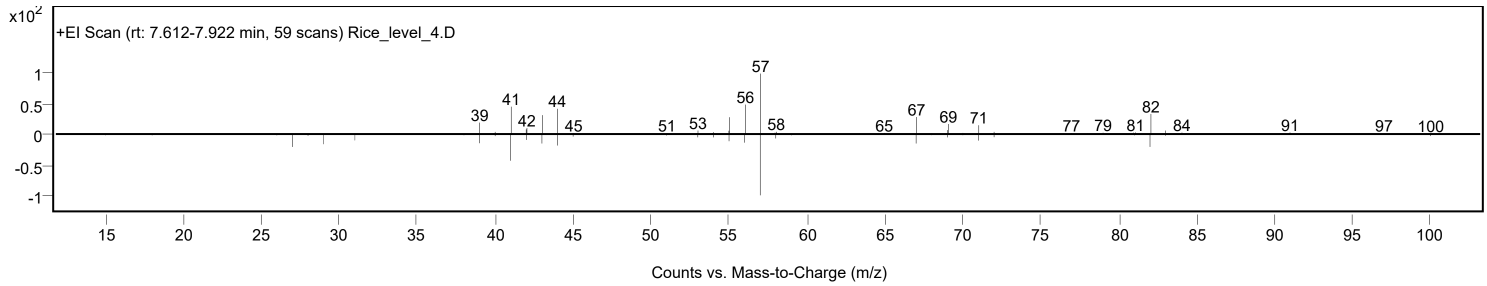
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	2-Hexen-1-ol, (E)-	C6H12O				928-95-0	65.40	65.40			NIST23.L
No	LibSearch	2-Hexen-1-ol, (Z)-	C6H12O				928-94-9	65.12	65.12			NIST23.L
No	LibSearch	Cyclopentanol, 2-methyl-	C6H12O				24070-77-7	64.33	64.33			NIST23.L
No	LibSearch	2-Hexen-1-ol, (E)-	C6H12O				928-95-0	64.30	64.30			NIST23.L
No	LibSearch	Cyclohexanol	C6H12O				108-93-0	64.25	64.25			NIST23.L
No	LibSearch	2-Hexen-1-ol, (Z)-	C6H12O				928-94-9	64.21	64.21			NIST23.L
No	LibSearch	2-Hexen-1-ol, (E)-	C6H12O				928-95-0	64.01	64.01			NIST23.L
No	LibSearch	2-Hexen-1-ol, (E)-	C6H12O				928-95-0	63.91	63.91			NIST23.L
No	LibSearch	Cyclopentanol, 2-methyl-, cis-	C6H12O				25144-05-2	63.90	63.90			NIST23.L
No	LibSearch	2-Hexen-1-ol, (E)-	C6H12O				928-95-0	63.88	63.88			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	2-Hexen-1-ol, (E)-	C6H12O		928-95-0	14.467		65.40	65.40			NIST23.L

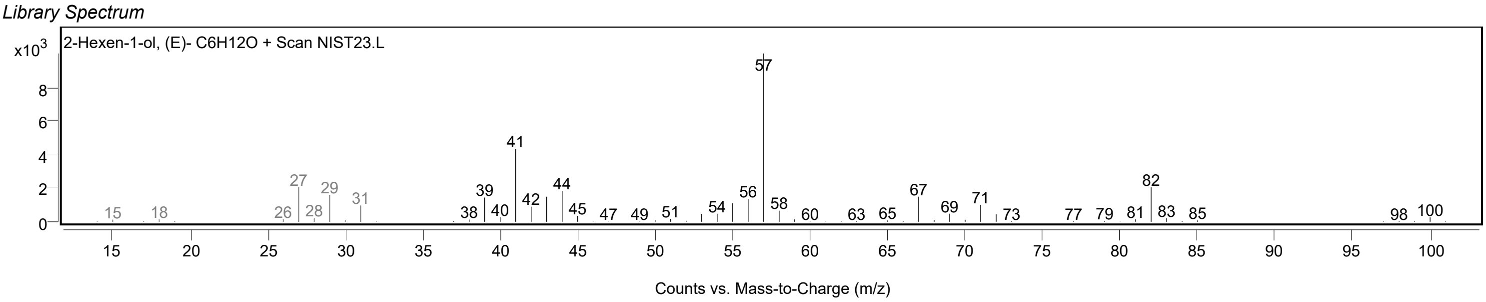
Observed Spectrum



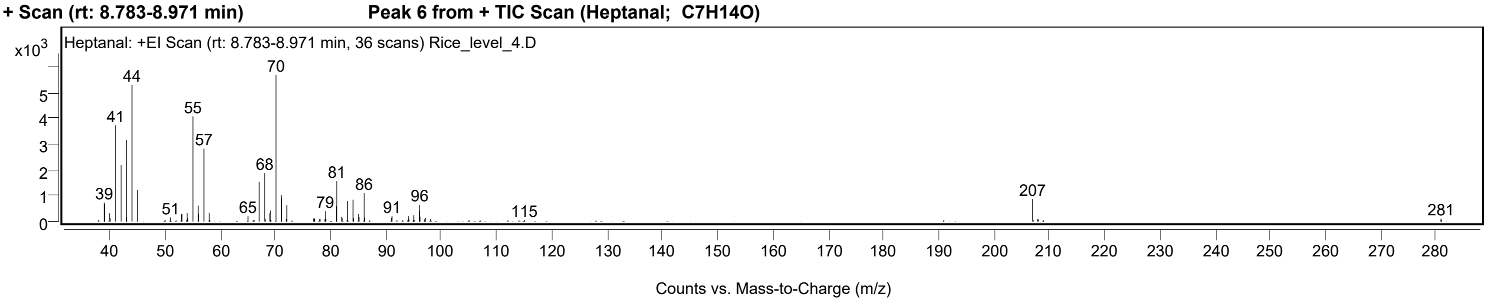
Mirror Plot



Library Spectrum



+ Scan (rt: 8.783-8.971 min) **Peak 6 from + TIC Scan (Heptanal; C7H14O)**



Spectrum Peaks

m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
37.9200		59	1.04					
38.9800		738	13.02					
39.0400		691	12.19					
39.9500		322	5.68					
40.0600		128	2.26					
41.0400		3721	65.67					
42.0400		2183	38.52					
42.9800		174	3.07					
43.0400		3146	55.51					
44.0100		5289	93.34					
45.0100		1230	21.71					
50.0100		66	1.17					
51.0000		160	2.83					
52.9500		303	5.35					
53.0500		285	5.02					
53.9000		65	1.15					
53.9900		333	5.88					
54.1000		121	2.13					
55.0400		4069	71.81					
55.9900		621	10.96					
56.0800		292	5.15					
57.0300		2824	49.84					
57.9800		344	6.06					
65.0000		209	3.68					
65.9900		59	1.05					
67.0200		1544	27.24					
68.0300		1889	33.33					
68.1300		113	1.99					
68.9500		331	5.84					
69.0400		430	7.58					
69.1500		69	1.21					
70.0700		5667	100.00					
71.0300		1021	18.01					
71.0800		923	16.28					
71.9200		102	1.80					
72.0300		630	11.12					
76.8500		117	2.07					
76.9700		124	2.18					
77.0900		112	1.98					
77.9400		110	1.94					
78.0600		91	1.60					
78.9100		86	1.52					
78.9900		400	7.06					
79.0800		113	2.00					
81.0100		605	10.67					
81.0700		1563	27.58					
81.9600		182	3.21					
82.0500		148	2.61					
82.9300		67	1.18					
83.0300		806	14.22					
84.0100		848	14.97					
84.1200		126	2.22					
84.9900		300	5.29					
85.1000		167	2.94					
86.0200		1106	19.51					
86.1000		154	2.72					
90.9500		140	2.47					
91.0600		214	3.77					
92.9400		59	1.04					
93.9400		74	1.31					
94.0200		205	3.61					
94.1300		92	1.62					
95.0100		244	4.31					
96.0100		408	7.20					
96.0700		651	11.49					
96.1400		191	3.38					
96.9900		111	1.97					
97.1000		147	2.60					
98.0500		96	1.70					
112.0500		58	1.02					
115.0000		64	1.12					
206.9800	1	877	15.47					
207.1100		58	1.03					
207.8400		89	1.57					
208.0100	1	95	1.68					
209.0000	1	67	1.19					
280.8800		115	2.03					
281.0100		72	1.27					

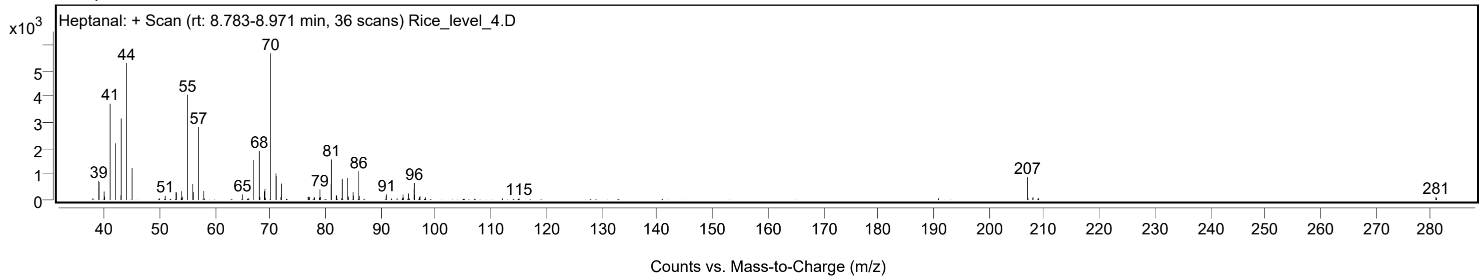
Spectrum Identification Table

Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	Heptanal	C7H14O				111-71-7	66.45	66.45			NIST23.L
No	LibSearch	Heptanal	C7H14O				111-71-7	66.42	66.42			NIST23.L
No	LibSearch	Heptanal	C7H14O				111-71-7	66.14	66.14			NIST23.L
No	LibSearch	Heptanal	C7H14O				111-71-7	65.61	65.61			NIST23.L
No	LibSearch	Heptanal	C7H14O				111-71-7	65.13	65.13			NIST23.L
No	LibSearch	Heptanal	C7H14O				111-71-7	65.04	65.04			NIST23.L
No	LibSearch	Cycloheptanol	C7H14O				502-41-0	59.51	59.51			NIST23.L
No	LibSearch	2-Hepten-1-ol, (E)-	C7H14O				33467-76-4	59.43	59.43			NIST23.L
No	LibSearch	2-Hepten-1-ol, (E)-	C7H14O				33467-76-4	58.96	58.96			NIST23.L
No	LibSearch	Cycloheptanol	C7H14O				502-41-0	58.43	58.43			NIST23.L

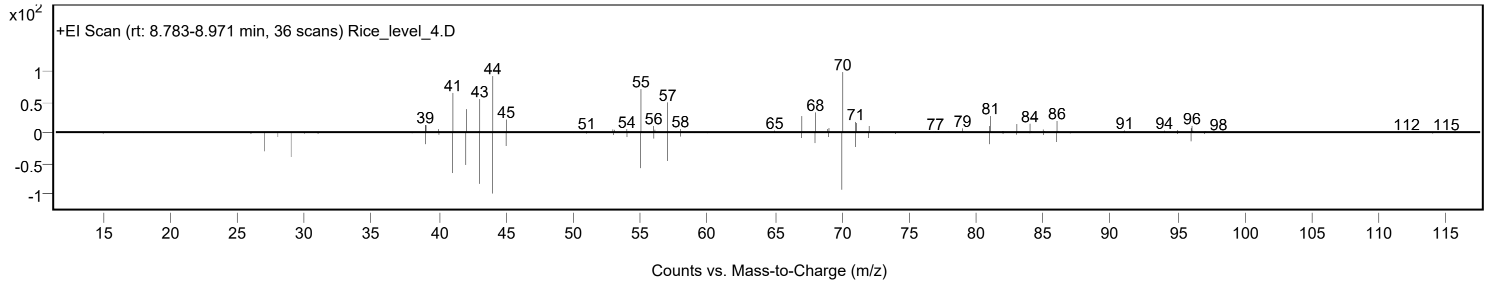
Analysis Report

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	Heptanal	C7H14O		111-71-7	15.100		66.45	66.45			NIST23.L

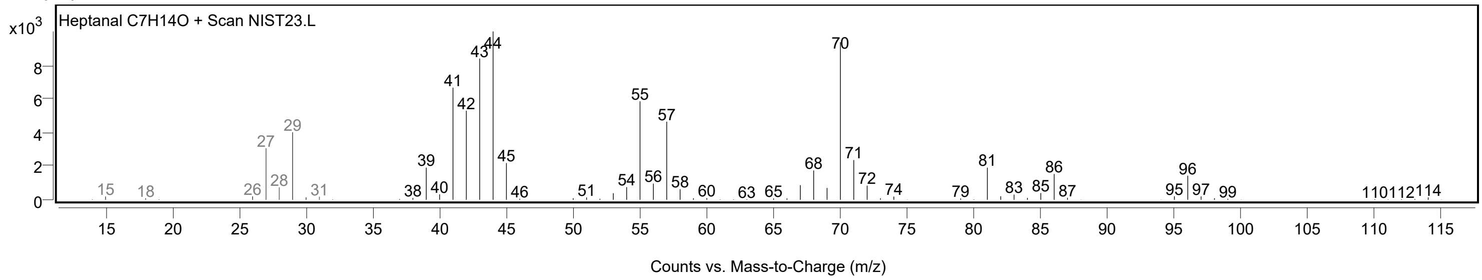
Observed Spectrum



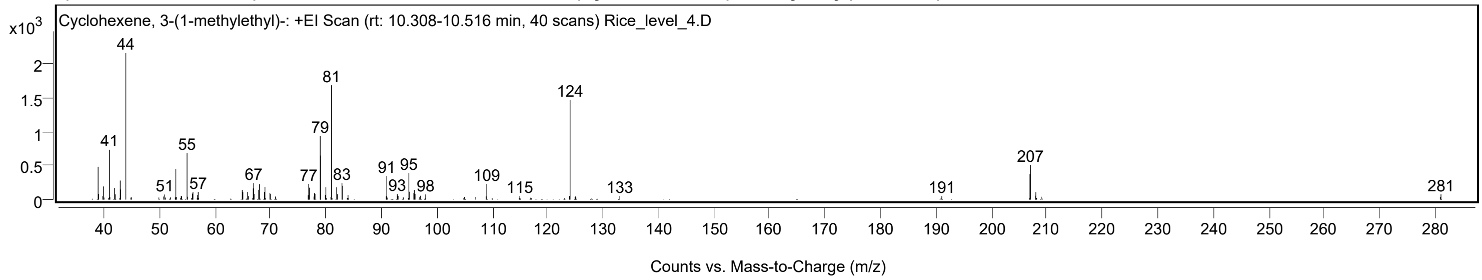
Mirror Plot



Library Spectrum



+ Scan (rt: 10.308-10.516 min) **Peak 7 from + TIC Scan (Cyclohexene, 3-(1-methylethyl)-; C9H16)**



Spectrum Peaks

m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
38.9800		486	22.44					
39.0800		85	3.94					
39.8300		51	2.36					
39.9600		194	8.99					
40.0500		31	1.43					
41.0200		737	34.05					
41.1100		36	1.67					
41.9400		172	7.95					
42.0600		77	3.58					
42.9700		281	13.01					
43.0600		149	6.87					
43.9800	1	2163	100.00					
44.8800		30	1.38					
45.0000	1	32	1.49					
49.9300		31	1.44					
50.8100		45	2.09					
50.9900		75	3.48					
51.1000		31	1.41					
52.0200		32	1.48					
52.8900		32	1.47					
52.9900		454	21.00					
53.0900		41	1.88					
53.9500		56	2.58					
54.0400		47	2.16					
55.0100		688	31.82					
55.1400		43	1.99					
55.9800		89	4.12					
56.0800		113	5.20					
56.8400		26	1.18					
56.9100		67	3.12					
57.0200		114	5.29					
57.1000		25	1.17					
64.9600		143	6.61					
65.0800		110	5.06					
65.9300		113	5.20					
66.0600		52	2.39					
66.9400		164	7.58					
67.0300		243	11.22					
67.1000		86	3.95					
67.9200		145	6.72					
68.0500		226	10.44					
68.9500		111	5.12					
69.0500		189	8.73					
69.9400		98	4.51					
70.0600		86	3.96					
70.9500		45	2.09					
76.8400		64	2.96					
76.9400		233	10.77					
77.0700		177	8.18					
77.8800		45	2.07					
77.9700		97	4.47					
78.0800		85	3.94					
79.0100	1	941	43.48					
79.0700		650	30.05					
79.9300	1	63	2.93					
80.0300		182	8.42					
80.9300	1	40	1.85					
81.0400	1	1688	78.05					
81.9800		181	8.36					
82.1000	1	80	3.68					
82.9300	1	245	11.30					
83.0500		205	9.47					
83.8900	1	26	1.21					
84.0200		70	3.25					
90.9100		48	2.22					
91.0100		346	15.98					
91.1700		30	1.38					
92.9000		79	3.66					
93.0500		56	2.58					
93.9800		32	1.48					
94.9900		392	18.10					
95.1200		117	5.39					
95.8800		98	4.53					
95.9700		144	6.65					
96.1100		86	3.99					
96.9400		35	1.63					
97.0600		57	2.66					
98.0200		76	3.53					
105.0100		41	1.90					
107.0500		41	1.91					
108.9200		47	2.19					
109.0300	1	233	10.76					
109.9900	1	24	1.09					
114.9800		52	2.39					
116.9400		28	1.30					
123.9400		51	2.35					
124.0400	1	1472	68.06					
124.9300		40	1.85					
125.0200	1	46	2.13					
125.1000		30	1.40					
133.0100		52	2.42					
191.0400		50	2.30					
206.9200	1	374	17.30					

Spectrum Peaks

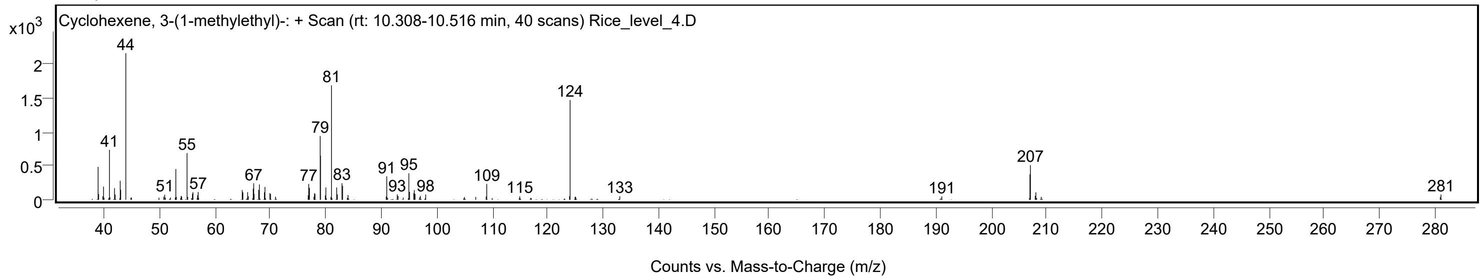
m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
207.0100	1	512	23.65					
207.8700	1	59	2.72					
207.9900	1	110	5.08					
208.1300		25	1.16					
208.9500	1	43	1.98					
280.8700		40	1.83					
281.0000		78	3.59					

Spectrum Identification Table

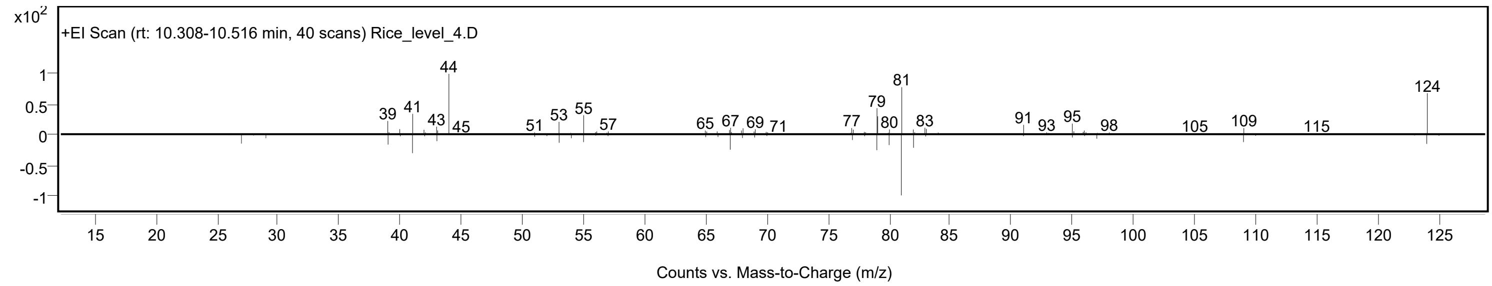
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	Cyclohexene, 3-(1-methylethyl)-	C9H16				3983-08-2	53.60	53.60			NIST23.L
No	LibSearch	1-Isopropylcyclohex-1-ene	C9H16				4292-04-0	53.26	53.26			NIST23.L
No	LibSearch	Cyclohexene, 3-(1-methylethyl)-	C9H16				3983-08-2	53.15	53.15			NIST23.L
No	LibSearch	Cyclohexane, (1-methylethylidene)-	C9H16				5749-72-4	52.69	52.69			NIST23.L
No	LibSearch	Cyclohexene,3-propyl-	C9H16				3983-06-0	51.46	51.46			NIST23.L
No	LibSearch	Cyclohexene,3-propyl-	C9H16				3983-06-0	51.45	51.45			NIST23.L
No	LibSearch	Cyclohexanone, 3-ethenyl-	C8H12O				1740-63-2	51.37	51.37			NIST23.L
No	LibSearch	Cyclohexene,1-propyl-	C9H16				2539-75-5	50.91	50.91			NIST23.L
No	LibSearch	Cyclohexane, (1-methylethylidene)-	C9H16				5749-72-4	50.75	50.75			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	Cyclohexene, 3-(1-methylethyl)-	C9H16		3983-08-2	15.283		53.60	53.60			NIST23.L

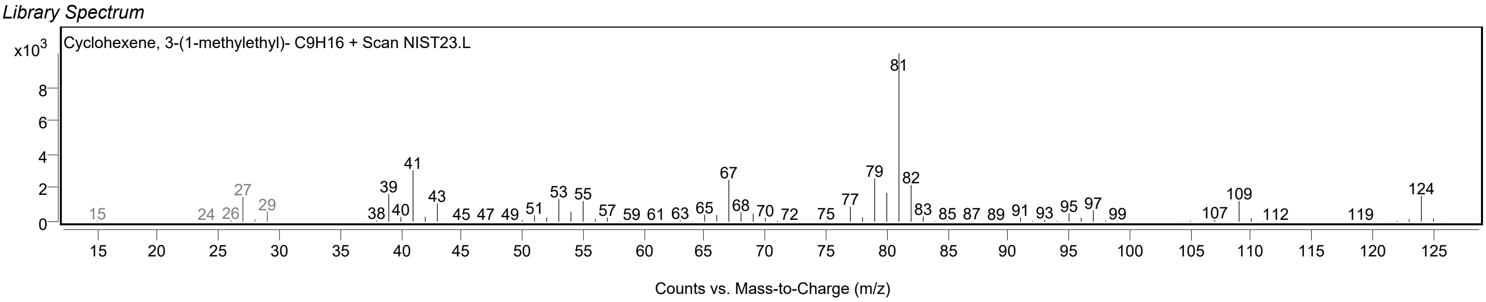
Observed Spectrum



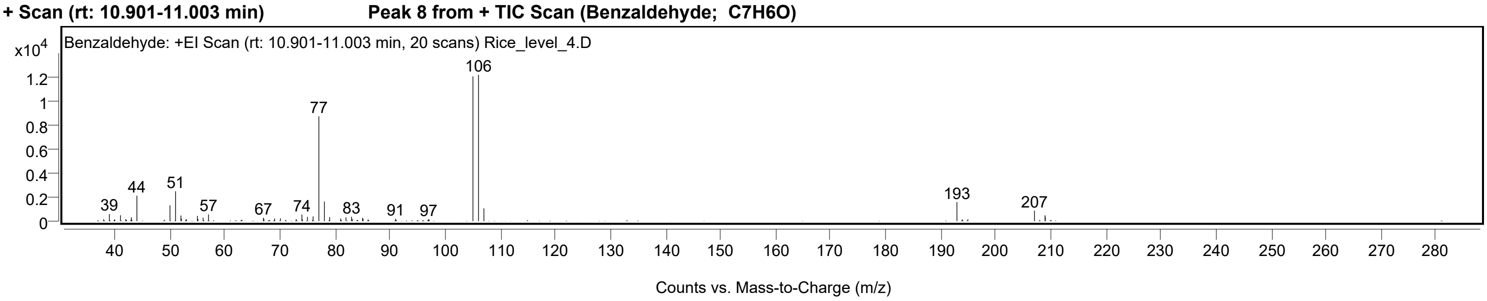
Mirror Plot



Library Spectrum



+ Scan (rt: 10.901-11.003 min) **Peak 8 from + TIC Scan (Benzaldehyde; C7H6O)**



Spectrum Peaks

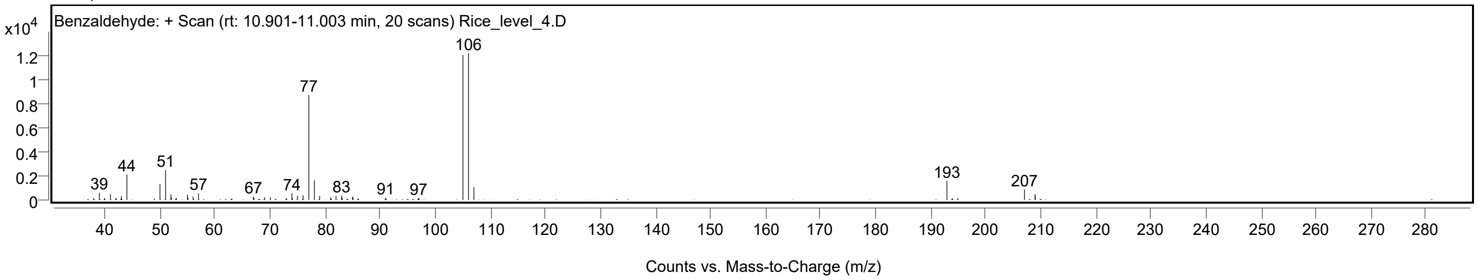
m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
37.9500		159	1.31					
38.9700		597	4.90					
39.9600		144	1.18					
41.0000		493	4.04					
41.9500		171	1.40					
42.9800		319	2.61					
43.9800		2118	17.38					
49.9900		1314	10.78					
51.0100		2493	20.46					
51.9800		469	3.85					
52.0700		204	1.68					
52.9800		157	1.29					
54.9700		435	3.57					
55.0700		182	1.49					
56.0100		288	2.36					
56.9900		563	4.62					
66.9600		280	2.30					
67.0700		160	1.31					
69.0100		224	1.84					
70.0500		232	1.91					
72.9200		162	1.33					
73.9500		554	4.55					
74.0500		139	1.14					
74.9500		357	2.93					
75.9900		395	3.24					
77.0300		8733	71.66					
78.0300		1637	13.43					
79.0000		342	2.81					
81.0000		213	1.75					
81.9600		319	2.62					
82.9700		364	2.99					
83.1000		167	1.37					
84.9600		276	2.26					
85.0700		188	1.54					
85.9700		126	1.04					
90.9800		204	1.68					
96.9700		159	1.30					
97.0700		123	1.01					
105.0200		12058	98.95					
106.0300	1	12187	100.00					
107.0100	1	1065	8.74					
192.9300	1	1588	13.03					
193.9400	1	132	1.08					
194.9200	1	155	1.27					
207.0100		875	7.18					
208.9300		485	3.98					
208.9900		357	2.93					

Spectrum Identification Table

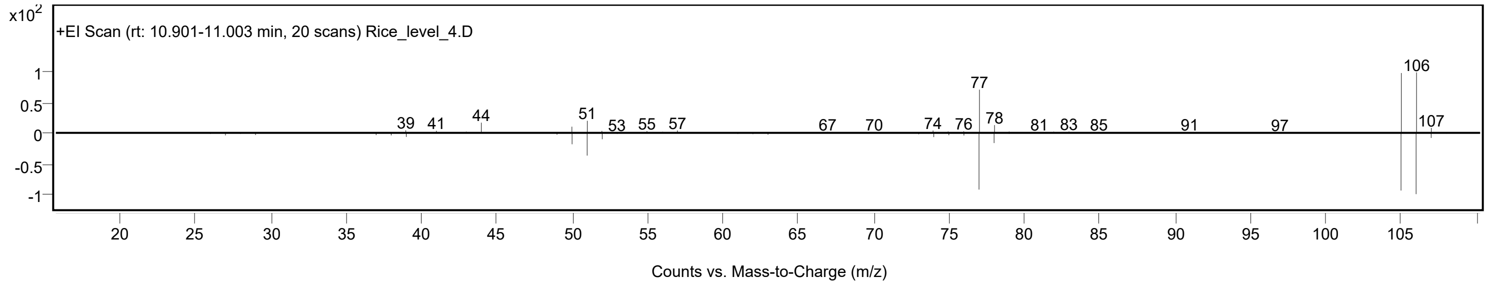
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	Benzaldehyde	C7H6O				100-52-7	60.92	60.92			NIST23.L
No	LibSearch	Benzaldehyde	C7H6O				100-52-7	59.80	59.80			NIST23.L
No	LibSearch	Benzaldehyde	C7H6O				100-52-7	59.26	59.26			NIST23.L
No	LibSearch	Benzaldehyde	C7H6O				100-52-7	58.99	58.99			NIST23.L
No	LibSearch	Benzaldehyde	C7H6O				100-52-7	58.85	58.85			NIST23.L
No	LibSearch	Benzaldehyde	C7H6O				100-52-7	58.38	58.38			NIST23.L
No	LibSearch	2,5-Pyrrolidinedione, 1-(benzoyloxy)-	C11H9NO4				23405-15-4	55.68	55.68			NIST23.L
No	LibSearch	Phenylglyoxal	C8H6O2				1074-12-0	55.23	55.23			NIST23.L
No	LibSearch	Phenacylidene diacetate	C12H12O5				5062-30-6	54.80	54.80			NIST23.L
No	LibSearch	Phenylglyoxylic acid, neopentyl ester	C13H16O3				1000453-45-9	54.23	54.23			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	Benzaldehyde	C7H6O		100-52-7	15.967		60.92	60.92			NIST23.L

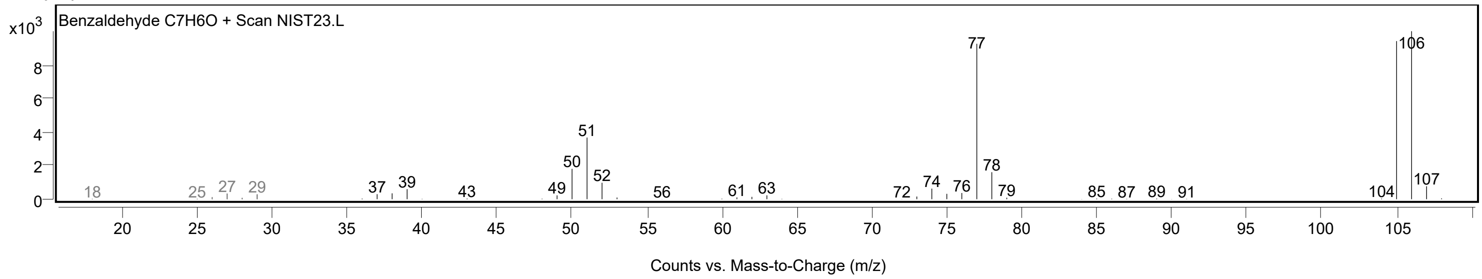
Observed Spectrum



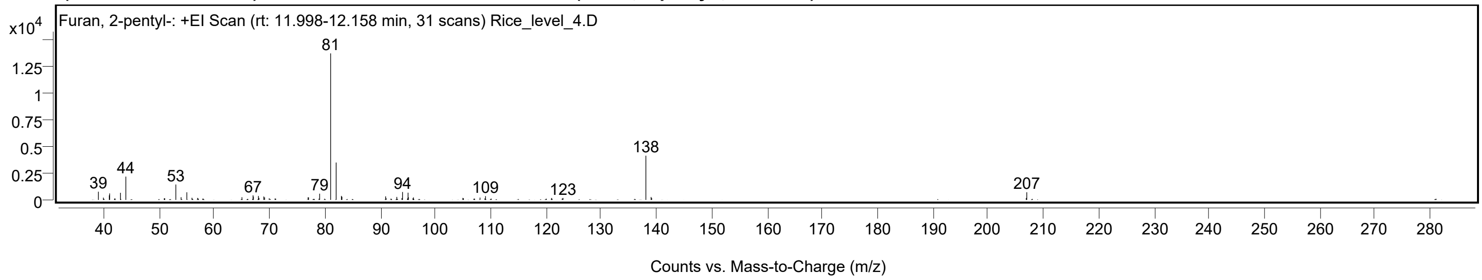
Mirror Plot



Library Spectrum



+ Scan (rt: 11.998-12.158 min) **Peak 9 from + TIC Scan (Furan, 2-pentyl-; C9H14O)**



Spectrum Peaks

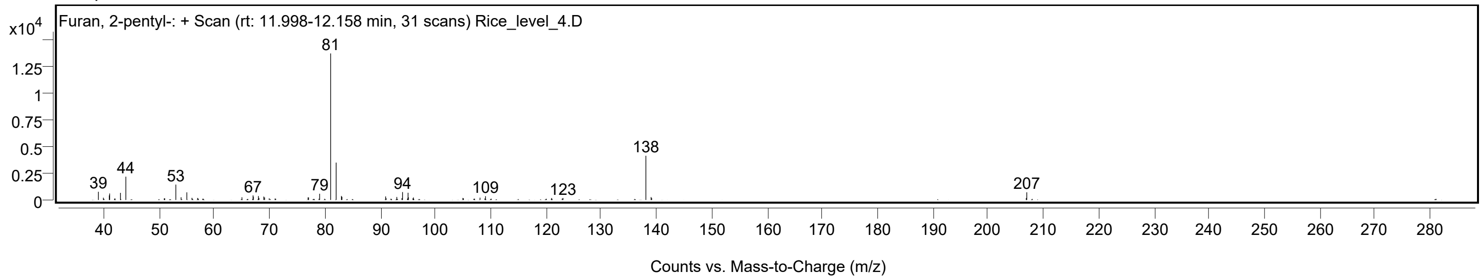
m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
39.0000		770	5.62					
39.9000		215	1.57					
40.9800		430	3.14					
41.0500		608	4.43					
42.0000		154	1.12					
42.9900		667	4.87					
43.9700		2179	15.91					
50.9200		149	1.09					
51.0200		149	1.09					
53.0200		1441	10.52					
53.9700		255	1.86					
55.0100		715	5.22					
55.9300		202	1.48					
56.9600		187	1.37					
64.9800		252	1.84					
66.9900		429	3.13					
67.0600		302	2.20					
67.9600		368	2.69					
68.0700		184	1.34					
68.9400		320	2.34					
69.0900		241	1.76					
69.9200		149	1.09					
76.9500		241	1.76					
77.0400		240	1.75					
79.0400		584	4.26					
81.0300		13700	100.00					
82.0300	1	3495	25.51					
82.9800		374	2.73					
83.0800	1	294	2.14					
90.9800		332	2.43					
91.0800		164	1.20					
93.0100		282	2.06					
93.9300		168	1.22					
94.0400		742	5.41					
95.0400		675	4.93					
95.1100		251	1.83					
95.9200		150	1.09					
96.0100		235	1.71					
104.9600		191	1.39					
105.0600		150	1.10					
108.0600		217	1.58					
108.9500		140	1.02					
109.0500		351	2.56					
121.0300		180	1.31					
123.0800		184	1.34					
138.0800	1	4136	30.19					
139.0100		269	1.97					
139.1300	1	174	1.27					
206.9000		182	1.33					
207.0000		720	5.25					

Spectrum Identification Table

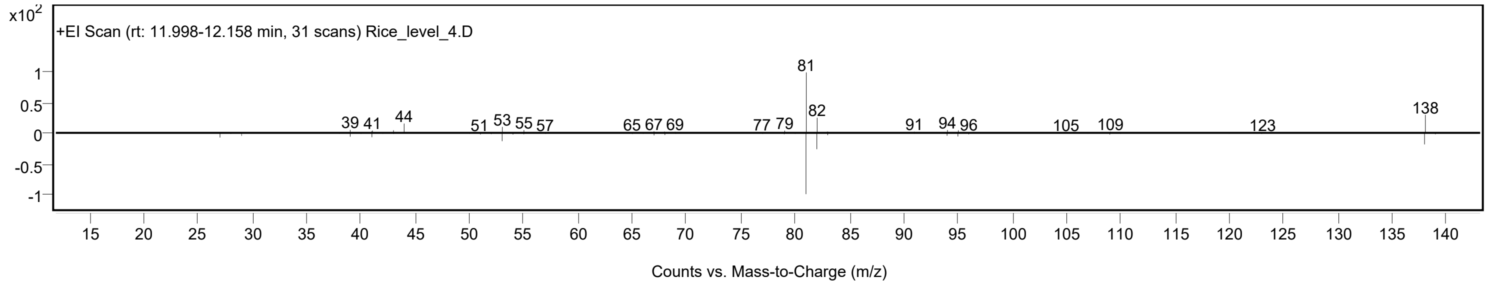
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	Furan, 2-pentyl-	C9H14O				3777-69-3	66.59	66.59			NIST23.L
No	LibSearch	Furan, 2-pentyl-	C9H14O				3777-69-3	64.21	64.21			NIST23.L
No	LibSearch	Furan, 2-pentyl-	C9H14O				3777-69-3	63.91	63.91			NIST23.L
No	LibSearch	Furan, 2-pentyl-	C9H14O				3777-69-3	61.82	61.82			NIST23.L
No	LibSearch	Furan, 2-pentyl-	C9H14O				3777-69-3	61.02	61.02			NIST23.L
No	LibSearch	2-[(2-Furylmethyl)amino]-2-methyl-1-propanol	C9H15NO2				889949-94-4	58.40	58.40			NIST23.L
No	LibSearch	2,4-Nonadienal, (E,E)-	C9H14O				5910-87-2	57.69	57.69			NIST23.L
No	LibSearch	2,4-Nonadienal, (E,E)-	C9H14O				5910-87-2	57.32	57.32			NIST23.L
No	LibSearch	9-Oxabicyclo[4.2.1]non-7-en-3-one	C8H10O2				76400-39-0	55.66	55.66			NIST23.L
No	LibSearch	2,4-Nonadienal, (E,E)-	C9H14O				5910-87-2	54.98	54.98			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	Furan, 2-pentyl-	C9H14O		3777-69-3	16.583		66.59	66.59			NIST23.L

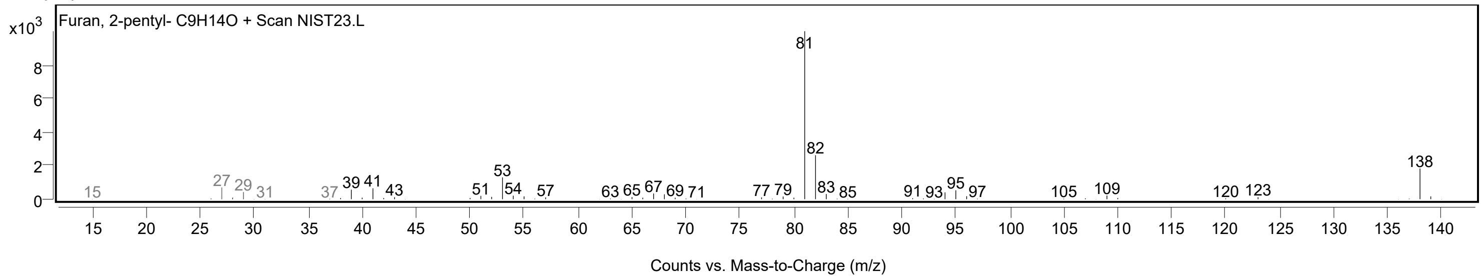
Observed Spectrum



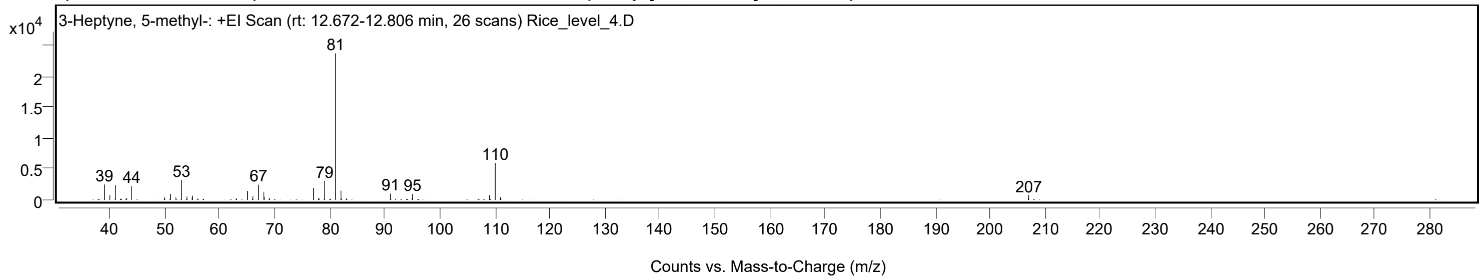
Mirror Plot



Library Spectrum



+ Scan (rt: 12.672-12.806 min) **Peak 10 from + TIC Scan (3-Heptyne, 5-methyl-; C8H14)**



Spectrum Peaks

m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
39.0200		2497	10.51					
40.0000		810	3.41					
41.0400		2409	10.14					
43.0100		301	1.27					
43.9700		2214	9.32					
49.9900		471	1.98					
51.0100		968	4.07					
52.0000		400	1.68					
53.0400		3242	13.64					
54.0200		549	2.31					
54.9600		374	1.58					
55.0400		691	2.91					
65.0100		1461	6.15					
66.0100		615	2.59					
67.0300		2475	10.42					
67.9900		1223	5.15					
68.0500		521	2.19					
68.9700		313	1.32					
77.0200		1960	8.25					
78.0200		289	1.22					
79.0500		3076	12.95					
81.0300	1	23758	100.00					
82.0100	1	1522	6.41					
91.0200		1017	4.28					
94.9500		249	1.05					
95.0300		960	4.04					
109.0000		770	3.24					
110.0500	1	5986	25.19					
111.0000	1	439	1.85					
207.0200		744	3.13					

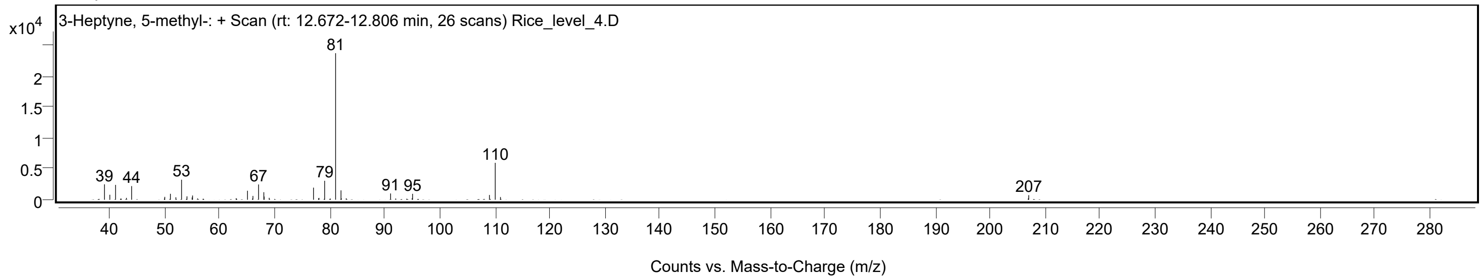
Analysis Report

Spectrum Identification Table

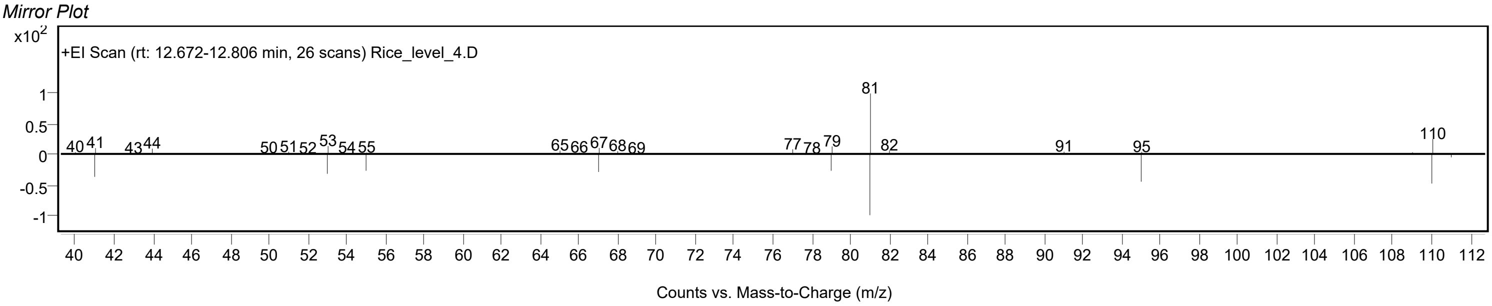
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	3-Heptyne, 5-methyl-	C8H14				61228-09-9	70.16	70.16			NIST23.L
No	LibSearch	2,4-Heptadienal, (E,E)-	C7H10O				4313-03-5	67.77	67.77			NIST23.L
No	LibSearch	2,4-Heptadienal, (E,E)-	C7H10O				4313-03-5	67.61	67.61			NIST23.L
No	LibSearch	2-Methyl-2,3-divinyloxirane	C7H10O				70597-50-1	65.49	65.49			NIST23.L
No	LibSearch	Cyclohexene, 1-ethyl-	C8H14				1453-24-3	64.30	64.30			NIST23.L
No	LibSearch	Cyclohexene, 3-ethyl-	C8H14				2808-71-1	64.21	64.21			NIST23.L
No	LibSearch	Cyclohexene, 1-ethyl-	C8H14				1453-24-3	64.20	64.20			NIST23.L
No	LibSearch	Cyclohexene, 1-ethyl-	C8H14				1453-24-3	64.14	64.14			NIST23.L
No	LibSearch	2,4-Octadiene	C8H14				13643-08-8	63.16	63.16			NIST23.L
No	LibSearch	1,4-Hexadiene, 3-ethyl-	C8H14				2080-89-9	63.02	63.02			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	3-Heptyne, 5-methyl-	C8H14		61228-09-9	12.750		70.16	70.16			NIST23.L

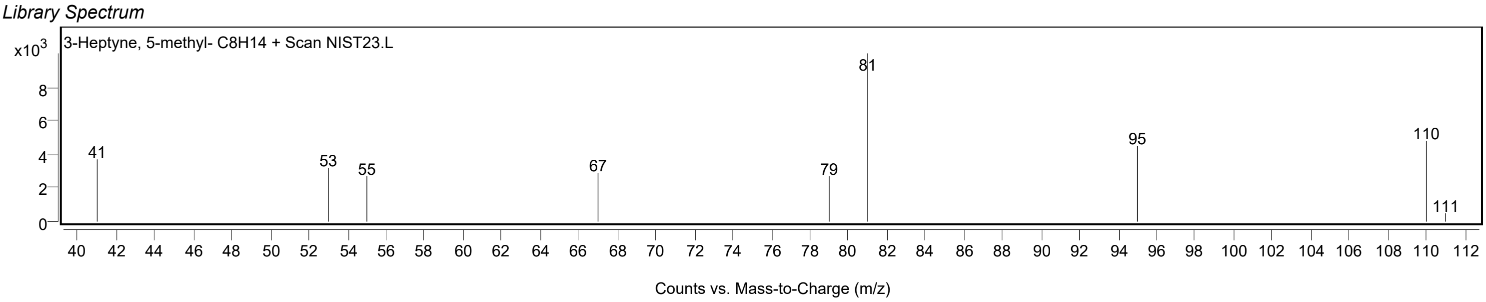
Observed Spectrum



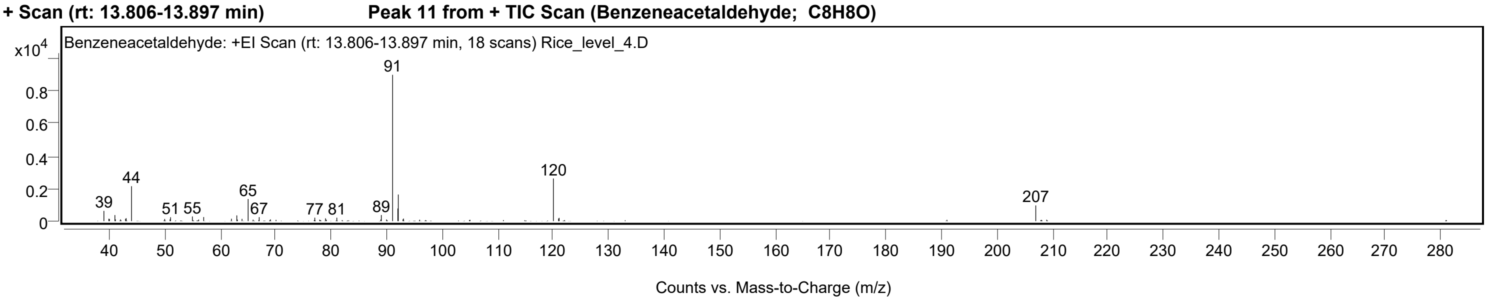
Mirror Plot



Library Spectrum



+ Scan (rt: 13.806-13.897 min) **Peak 11 from + TIC Scan (Benzeneacetaldehyde; C8H8O)**



Spectrum Peaks

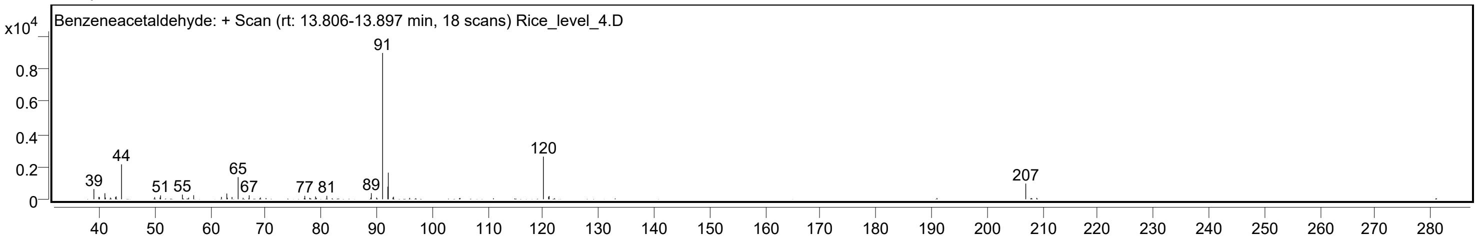
m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
39.0000		629	7.00					
39.8900		146	1.63					
40.0000		133	1.48					
40.9800		371	4.13					
41.0800		106	1.17					
41.9900		104	1.16					
42.9000		139	1.55					
43.0200		179	1.99					
43.9800		2147	23.89					
49.9600		132	1.47					
50.9200		104	1.16					
51.0200		240	2.67					
54.9500		286	3.19					
56.0900		96	1.07					
56.9900		250	2.78					
61.9900		147	1.63					
62.9600		351	3.91					
63.0600		92	1.02					
63.9000		139	1.54					
65.0000		1360	15.14					
67.0000		246	2.74					
69.0200		106	1.18					
77.0200		212	2.36					
78.9500		171	1.90					
80.9900		212	2.36					
88.9200		97	1.08					
89.0100		375	4.17					
89.9500		101	1.13					
91.0500	1	8987	100.00					
91.9900	1	777	8.64					
92.0700		1635	18.20					
92.9100		104	1.16					
93.0100	1	155	1.73					
120.0300	1	2619	29.14					
120.9600		156	1.73					
121.0700	1	202	2.25					
206.9900		965	10.74					

Spectrum Identification Table

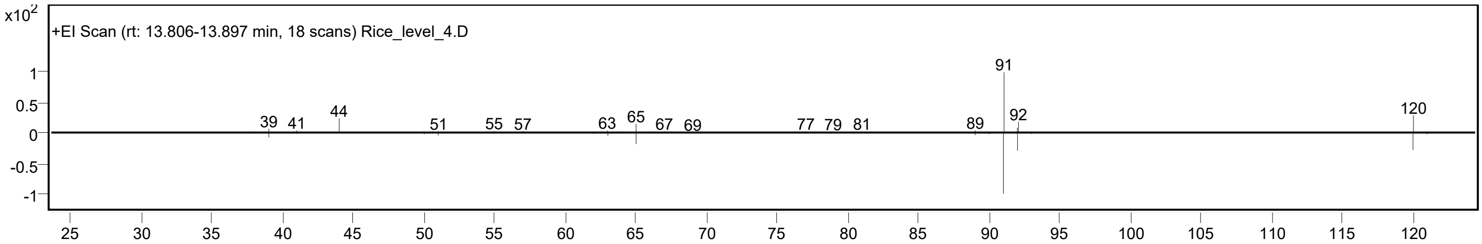
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	Benzeneacetaldehyde	C8H8O				122-78-1	60.89	60.89			NIST23.L
No	LibSearch	N-Isobutyl(phenyl)methanesulfonamide	C11H17NO2S				144615-94-1	59.32	59.32			NIST23.L
No	LibSearch	Benzeneacetaldehyde	C8H8O				122-78-1	58.17	58.17			NIST23.L
No	LibSearch	Benzeneacetaldehyde	C8H8O				122-78-1	58.06	58.06			NIST23.L
No	LibSearch	Benzeneacetaldehyde	C8H8O				122-78-1	57.83	57.83			NIST23.L
No	LibSearch	Benzeneacetaldehyde	C8H8O				122-78-1	56.54	56.54			NIST23.L
No	LibSearch	2,4,6-Cycloheptatrien-1-one, 4-methyl-	C8H8O				1000327-34-9	55.95	55.95			NIST23.L
No	LibSearch	Benzeneacetaldehyde	C8H8O				122-78-1	55.82	55.82			NIST23.L
No	LibSearch	1-(Benzylamino)propan-2-ol	C10H15NO				27159-32-6	55.57	55.57			NIST23.L
No	LibSearch	Benzeneacetaldehyde	C8H8O				122-78-1	55.51	55.51			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	Benzeneacetaldehyde	C8H8O		122-78-1	17.550		60.89	60.89			NIST23.L

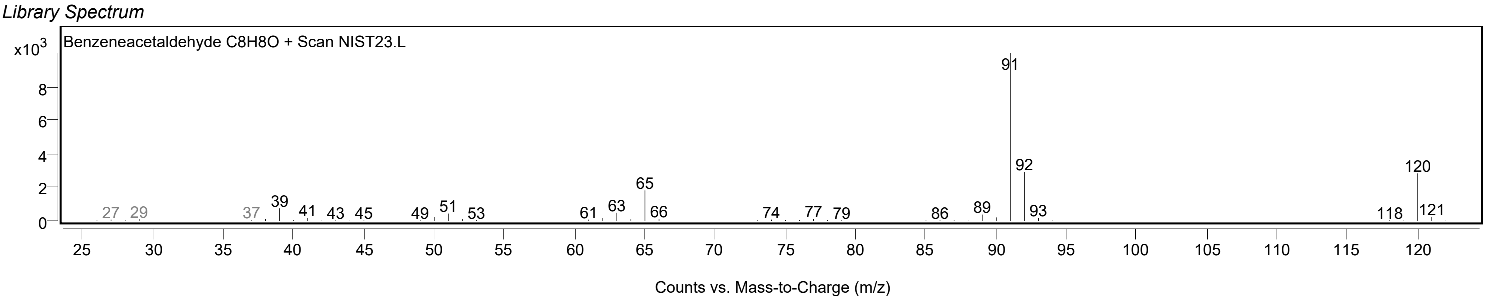
Observed Spectrum



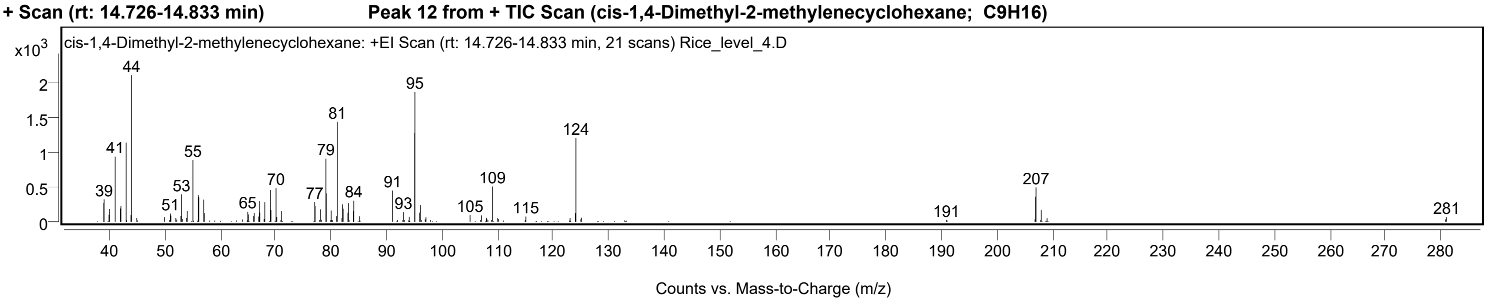
Mirror Plot



Library Spectrum



+ Scan (rt: 14.726-14.833 min) **Peak 12 from + TIC Scan (cis-1,4-Dimethyl-2-methylenecyclohexane; C9H16)**



Spectrum Peaks

m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
38.9500		274	12.93					
39.0200		323	15.24					
39.9000		102	4.80					
40.0000		187	8.81					
41.0100		941	44.45					
41.9500		191	9.00					
42.0500		229	10.82					
42.9900		1144	54.02					
43.8600		96	4.53					
43.9700	1	2117	100.00					
44.9100	1	56	2.64					
49.9300		67	3.15					
50.8300		24	1.16					
50.9600		118	5.58					
51.0600		88	4.16					
51.9400		54	2.55					
52.1200		31	1.45					
52.8600		83	3.94					
52.9900		395	18.65					
53.1300		39	1.82					
53.8600		57	2.69					
54.0000		157	7.41					
55.0300		891	42.07					
55.9900		388	18.31					
56.0900		357	16.87					
56.9800		320	15.11					
57.0800		126	5.96					
63.9300		32	1.51					
64.9200		144	6.82					
65.0300		105	4.95					
65.9100		84	3.95					
66.0800		125	5.92					
66.8900		74	3.51					
66.9900		298	14.06					
67.0900		132	6.23					
67.8600		35	1.64					
67.9900		281	13.28					
68.9900		461	21.75					
69.1000		167	7.90					
70.0100	1	486	22.94					
70.1500		68	3.21					
70.8800	1	21	1.01					
71.0300		157	7.39					
76.9900		286	13.50					
77.0600		231	10.91					
77.8500		26	1.24					
78.0200		178	8.40					
78.2200		30	1.40					
79.0000	1	912	43.07					
79.0800		409	19.31					
79.8500	1	24	1.14					
79.9500		162	7.64					
80.0900		25	1.18					
80.9200	1	81	3.83					
81.0500		1446	68.27					
81.8900		52	2.44					
81.9900		250	11.83					
82.1100		189	8.95					
82.9700		130	6.15					
83.0800		269	12.73					
83.1800		24	1.15					
83.9600		151	7.11					
84.0300		305	14.39					
85.0200		79	3.75					
91.0200	1	451	21.29					
91.9300	1	28	1.33					
93.0000	1	139	6.59					
93.1800		24	1.12					
94.0200		72	3.41					
95.0000		1281	60.49					
95.0500	1	1877	88.66					
95.9100	1	132	6.24					
96.0500		239	11.29					
96.2000		25	1.20					
96.9000		33	1.55					
97.0400	1	64	3.03					
97.8700		24	1.14					
104.9900		97	4.58					
106.8900		26	1.21					
107.0500		92	4.32					
107.9300		58	2.73					
108.0300		35	1.65					
108.8600		27	1.27					
109.0200	1	509	24.05					
109.9800	1	56	2.64					
110.1100		40	1.89					
115.0500		75	3.54					
123.0200		57	2.69					
123.9600		123	5.83					
124.0700	1	1211	57.19					
124.9600		35	1.67					
125.0700	1	60	2.81					
190.8500		27	1.26					

Spectrum Peaks

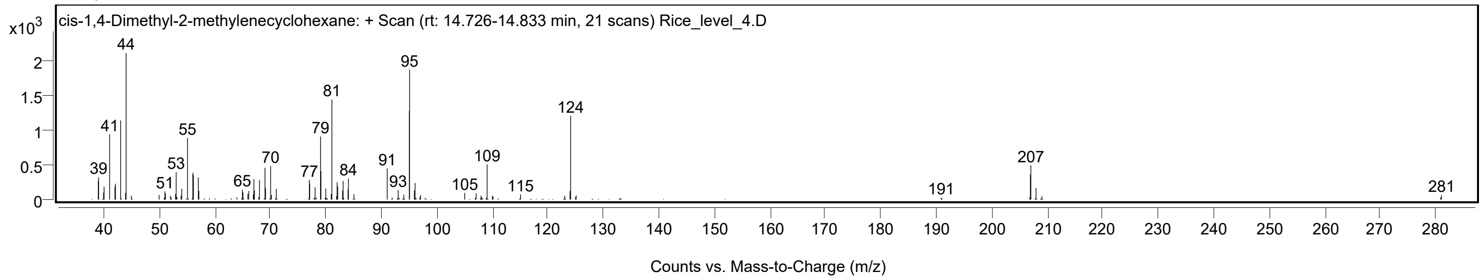
m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
206.8300		34	1.60					
206.9200		362	17.10					
207.0200		496	23.41					
207.9500		169	8.00					
209.0100		52	2.44					
280.8400		33	1.57					
281.0300		66	3.12					

Spectrum Identification Table

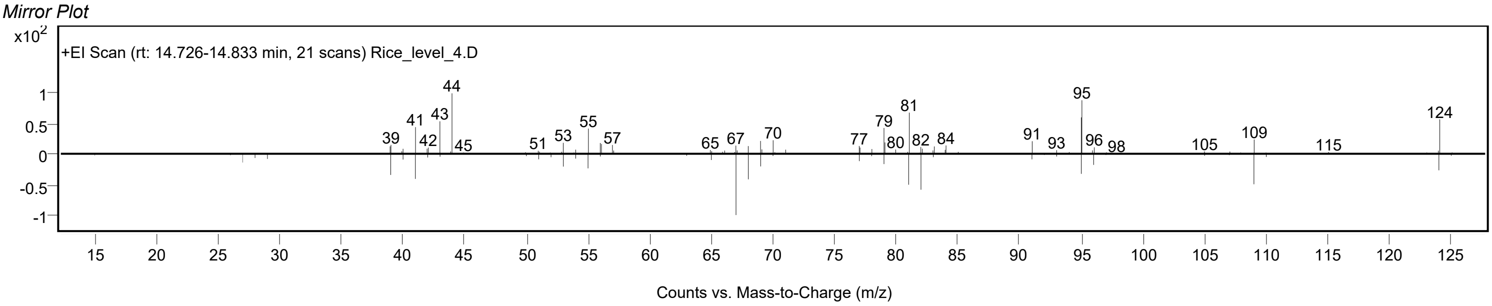
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	cis-1,4-Dimethyl-2-methylenecyclohexane	C9H16				19781-46-5	66.12	66.12			NIST23.L
No	LibSearch	3,4-Octadiene, 7-methyl-	C9H16				37050-05-8	60.26	60.26			NIST23.L
No	LibSearch	Cyclohexane, 1,3-dimethyl-2-methylene-, trans-	C9H16				20348-74-7	59.28	59.28			NIST23.L
No	LibSearch	Cyclohexane, 1,3-dimethyl-2-methylene-, cis-	C9H16				19781-47-6	59.02	59.02			NIST23.L
No	LibSearch	1-Nonyne	C9H16				3452-09-3	58.49	58.49			NIST23.L
No	LibSearch	1-Nonyne	C9H16				3452-09-3	58.35	58.35			NIST23.L
No	LibSearch	1,3-Hexadiene, 3-ethyl-2-methyl-, (Z)-	C9H16				74752-97-9	58.27	58.27			NIST23.L
No	LibSearch	Cyclopentene,1-(2-methylpropyl)-	C9H16				53098-47-8	56.85	56.85			NIST23.L
No	LibSearch	3-Octyne, 7-methyl-	C9H16				37050-06-9	56.72	56.72			NIST23.L
No	LibSearch	3-Cyclohexene-1-carboxaldehyde, 1-methyl-	C8H12O				931-96-4	56.49	56.49			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	cis-1,4-Dimethyl-2-methylenecyclohexane	C9H16		19781-46-5	14.750		66.12	66.12			NIST23.L

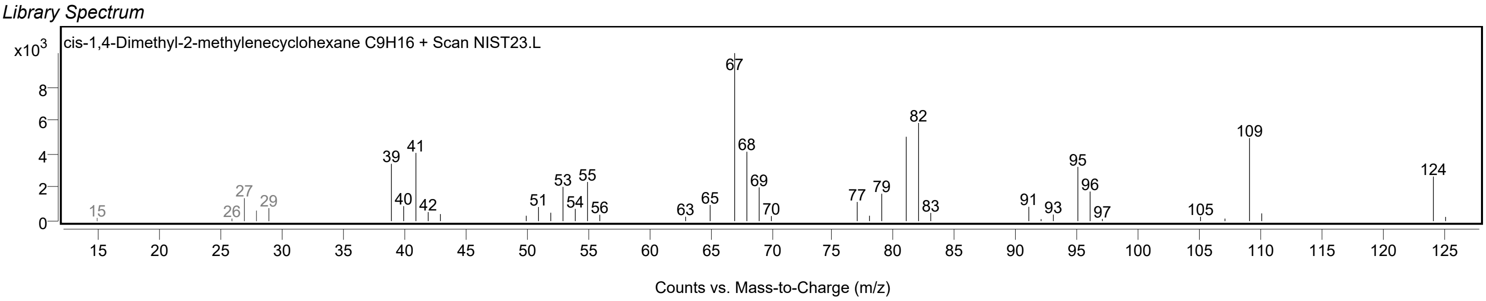
Observed Spectrum



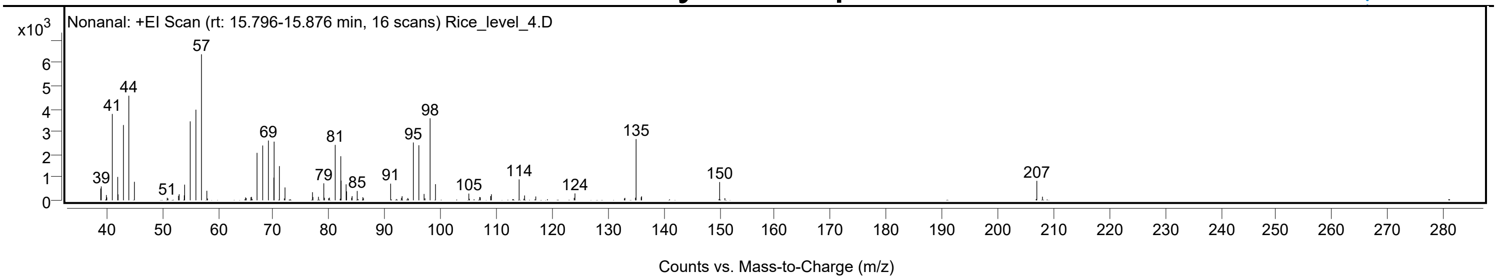
Mirror Plot



Library Spectrum



+ Scan (rt: 15.796-15.876 min) **Peak 13 from + TIC Scan (Nonanal; C9H18O)**


Nonanal: +EI Scan (rt: 15.796-15.876 min, 16 scans) Rice_level_4.D
x10^3
57
44
41
98
69
81
95
135
39
51
79
85
91
105
114
124
150
207
Counts vs. Mass-to-Charge (m/z)

Spectrum Peaks

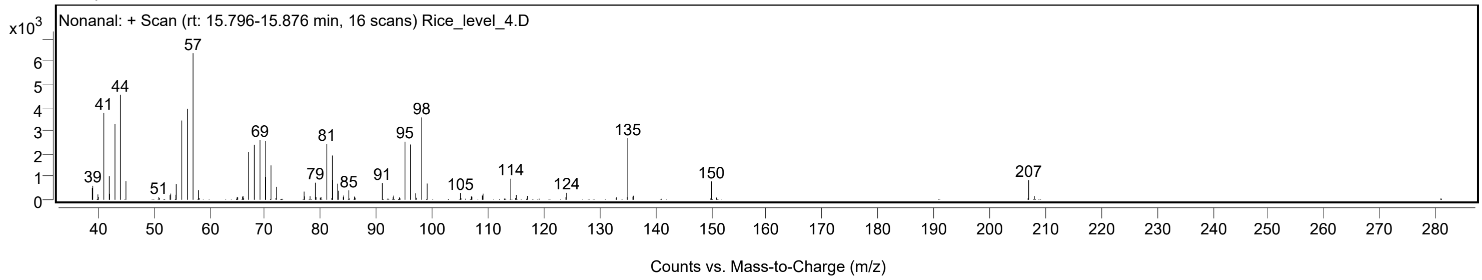
m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
38.9700		515	8.08					
39.0600		608	9.53					
39.9800		231	3.62					
40.0600		127	2.00					
41.0400		3770	59.14					
42.0200		1021	16.02					
42.0800		260	4.08					
43.0400		3287	51.56					
44.0100		4568	71.65					
45.0100		809	12.69					
50.9000		110	1.73					
51.0400		83	1.30					
52.9200		198	3.11					
53.0500		267	4.19					
53.9600		217	3.41					
54.0300		685	10.75					
55.0300		3447	54.06					
56.0500		3956	62.06					
57.0500	1	6375	100.00					
58.0100	1	416	6.52					
58.1400		80	1.26					
64.9600		127	1.99					
65.0800		99	1.56					
65.9300		136	2.13					
66.0300		150	2.35					
67.0200		2073	32.52					
68.0400		2394	37.55					
69.0600		2610	40.95					
70.0400		992	15.56					
70.0900		2562	40.19					
71.0300	1	1494	23.44					
71.9100	1	82	1.29					
72.0400		564	8.85					
76.9700		354	5.56					
77.0600		144	2.26					
78.0400		154	2.41					
79.0000		742	11.64					
79.1200		113	1.77					
79.8700		90	1.41					
80.0100		118	1.85					
81.0500	1	2419	37.95					
81.9400	1	69	1.08					
82.0500		1924	30.18					
82.1100		860	13.49					
83.0100	1	700	10.97					
83.1000		393	6.16					
84.0000		111	1.73					
84.0900		184	2.88					
85.0000		408	6.40					
86.0100		137	2.14					
86.1300		89	1.40					
91.0000		728	11.43					
92.9300		106	1.66					
93.0600		180	2.82					
94.0800		100	1.57					
95.0700		2526	39.62					
96.0600	1	2403	37.69					
97.0000	1	278	4.36					
97.1700		87	1.37					
98.0800		3581	56.17					
99.0500		705	11.06					
105.0300		295	4.63					
105.1300		65	1.02					
106.9800		144	2.26					
107.0900		132	2.06					
108.9500		184	2.88					
109.0800		265	4.16					
114.0600	1	915	14.35					
114.9100	1	68	1.06					
115.0400		213	3.34					
117.0500		169	2.65					
123.9400		109	1.71					
124.0800		302	4.74					
132.9400		81	1.27					
133.0500		106	1.67					
134.9300		116	1.82					
135.0500	1	2674	41.95					
135.9600		132	2.08					
136.0400	1	175	2.75					
150.0500	1	799	12.53					
150.9800	1	95	1.50					
206.9700	1	851	13.35					
207.9800	1	160	2.50					

Spectrum Identification Table

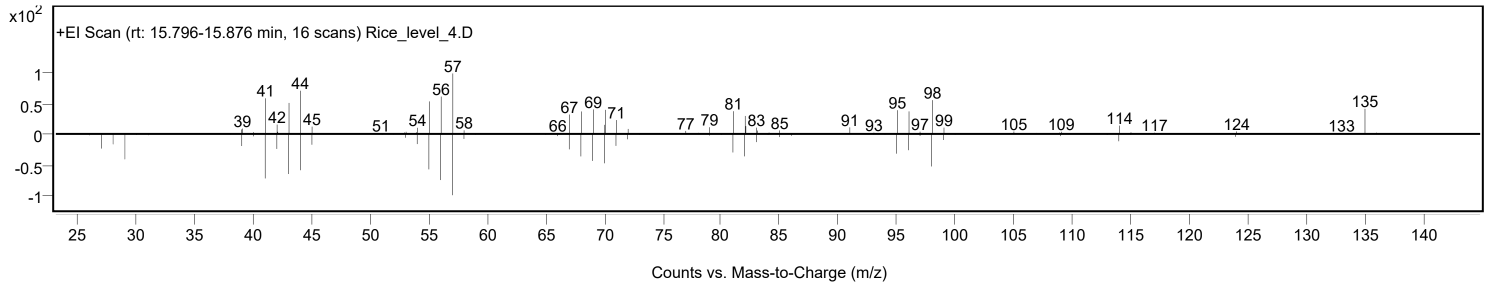
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	Nonanal	C9H18O				124-19-6	65.12	65.12			NIST23.L
No	LibSearch	Nonanal	C9H18O				124-19-6	63.87	63.87			NIST23.L
No	LibSearch	Nonanal	C9H18O				124-19-6	63.54	63.54			NIST23.L
No	LibSearch	Nonanal	C9H18O				124-19-6	63.35	63.35			NIST23.L
No	LibSearch	Nonanal	C9H18O				124-19-6	63.12	63.12			NIST23.L
No	LibSearch	Cyclopentane, 1-methyl-3-(2-methylpropyl)-	C10H20				29053-04-1	62.54	62.54			NIST23.L
No	LibSearch	Cyclopropane, hexylidene-	C9H16				91509-06-7	61.44	61.44			NIST23.L
No	LibSearch	4-Methyl-cyclohex-2-en-1-ol	C7H12O				1000144-17-5	59.40	59.40			NIST23.L
No	LibSearch	2-Nonen-1-ol, (E)-	C9H18O				31502-14-4	58.24	58.24			NIST23.L
No	LibSearch	2-Nonen-1-ol, (E)-	C9H18O				31502-14-4	58.23	58.23			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	Nonanal	C9H18O		124-19-6	18.467		65.12	65.12			NIST23.L

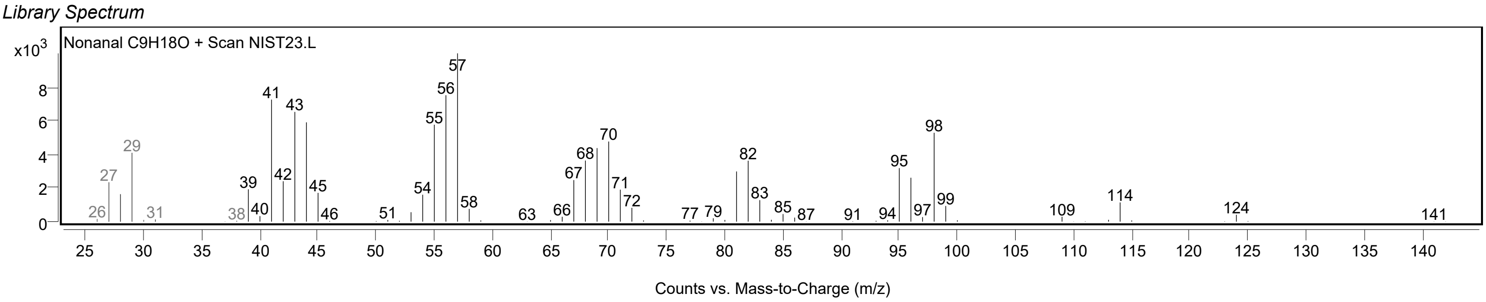
Observed Spectrum



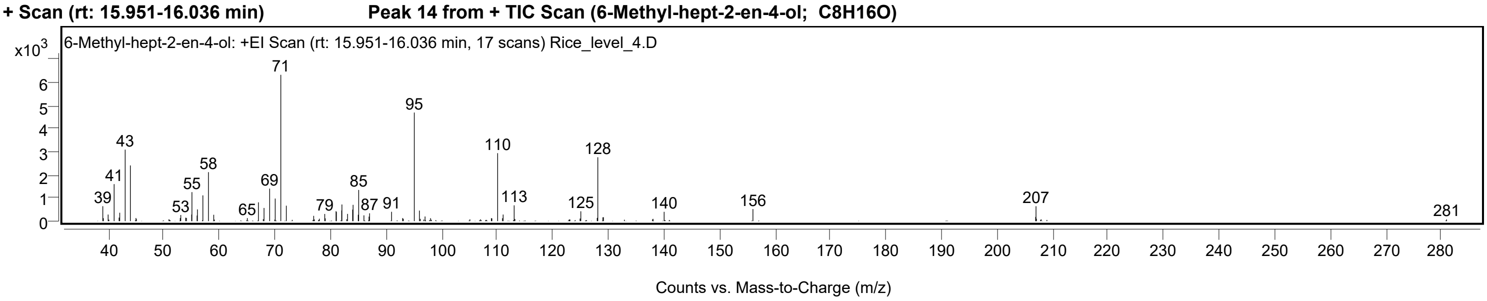
Mirror Plot



Library Spectrum



+ Scan (rt: 15.951-16.036 min) **Peak 14 from + TIC Scan (6-Methyl-hept-2-en-4-ol; C8H16O)**



Spectrum Peaks

m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
38.9900		645	10.20					
39.0900		126	2.00					
39.9500		284	4.50					
41.0300		1599	25.31					
41.9200		152	2.40					
42.0400		362	5.73					
43.0200		3088	48.87					
43.9500		2405	38.05					
44.9600		133	2.11					
45.0500		83	1.32					
50.8900		78	1.24					
52.9300		132	2.08					
53.0300		265	4.19					
53.9100		148	2.34					
54.0600		139	2.19					
54.9800		285	4.50					
55.0400		1253	19.83					
55.9600		242	3.83					
56.0300		506	8.00					
57.0100		1112	17.60					
58.0100		2116	33.48					
58.9600		274	4.34					
59.0800		65	1.02					
65.0100		131	2.08					
67.0100		812	12.86					
67.8800		84	1.33					
68.0200		569	9.00					
69.0300		1405	22.23					
70.0300		975	15.43					
71.0400		6319	100.00					
72.0300		667	10.56					
76.9600		229	3.62					
77.0900		72	1.15					
77.9800		103	1.63					
78.9500		310	4.90					
79.0400		120	1.89					
80.9800		429	6.78					
81.0500		397	6.28					
81.9900		462	7.31					
82.0400		718	11.37					
82.9100		80	1.26					
83.0600		312	4.93					
83.9700		498	7.88					
84.0500		703	11.13					
84.9800		274	4.34					
85.0600		1347	21.31					
86.0000		250	3.96					
86.9200		194	3.08					
87.0200		339	5.36					
90.9800		404	6.39					
92.9700		130	2.06					
93.0900		101	1.60					
95.0700	1	4690	74.22					
95.9900		455	7.20					
96.1500	1	98	1.54					
96.7800		67	1.07					
97.0100	1	201	3.18					
97.9900		127	2.01					
105.0800		77	1.21					
106.9800		77	1.22					
108.9400		127	2.01					
109.0400		118	1.87					
110.0800	1	2935	46.44					
110.9400		126	1.99					
111.0700	1	285	4.51					
113.0500		681	10.78					
113.1900		94	1.48					
123.0100		74	1.17					
123.0900		70	1.10					
124.9100		111	1.76					
125.0500		425	6.72					
128.0000		140	2.21					
128.1100	1	2759	43.66					
129.0200		165	2.61					
129.1300	1	168	2.66					
137.9500		73	1.15					
138.0600		98	1.55					
140.0500		401	6.35					
156.0400		522	8.26					
206.9300		182	2.88					
207.0000	1	643	10.18					
207.0800		101	1.60					
207.9400	1	74	1.17					
280.9200		73	1.15					

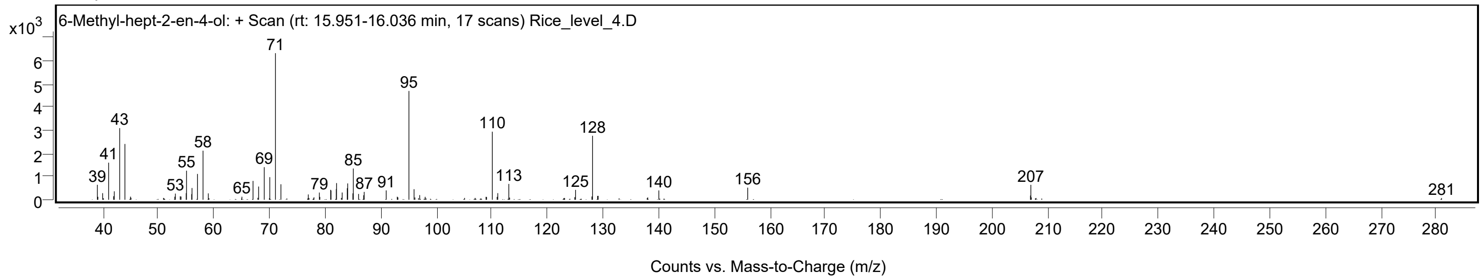
Analysis Report

Spectrum Identification Table

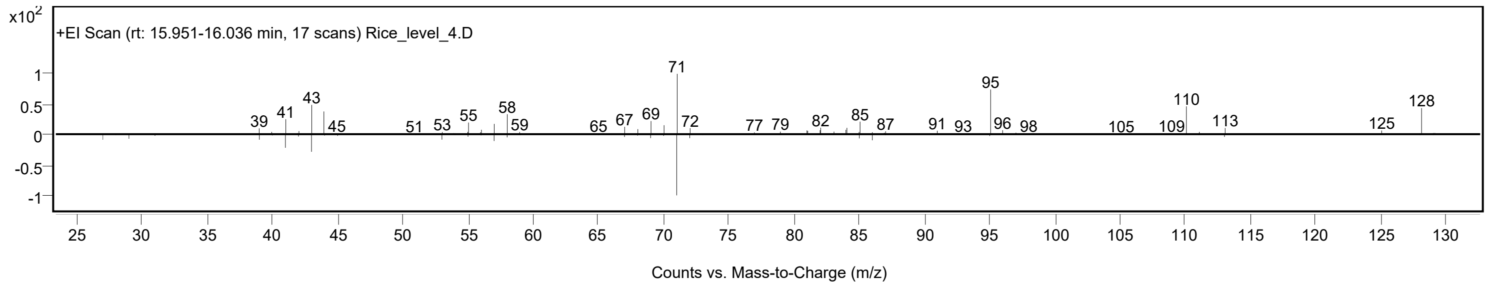
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	6-Methyl-hept-2-en-4-ol	C8H16O				83212-30-0	68.62	68.62			NIST23.L
No	LibSearch	Cyclohexanol, 2,6-dimethyl-	C8H16O				5337-72-4	60.12	60.12			NIST23.L
No	LibSearch	2,5-Dimethylcyclohexanol	C8H16O				3809-32-3	58.75	58.75			NIST23.L
No	LibSearch	2,5-Dimethylcyclohexanol	C8H16O				3809-32-3	58.30	58.30			NIST23.L
No	LibSearch	Cyclohexanol, 3,5-dimethyl-	C8H16O				5441-52-1	57.67	57.67			NIST23.L
No	LibSearch	Cyclohexanol, 3,5-dimethyl-	C8H16O				5441-52-1	57.46	57.46			NIST23.L
No	LibSearch	Cyclohexanol, 2,6-dimethyl-	C8H16O				5337-72-4	57.15	57.15			NIST23.L
No	LibSearch	3,4-Dimethylcyclohexanol	C8H16O				5715-23-1	56.22	56.22			NIST23.L
No	LibSearch	5-Hepten-2-ol, 6-methyl-	C8H16O				1569-60-4	56.02	56.02			NIST23.L
No	LibSearch	2-Heptanone, 6-methyl-	C8H16O				928-68-7	55.95	55.95			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	6-Methyl-hept-2-en-4-ol	C8H16O		83212-30-0	16.000		68.62	68.62			NIST23.L

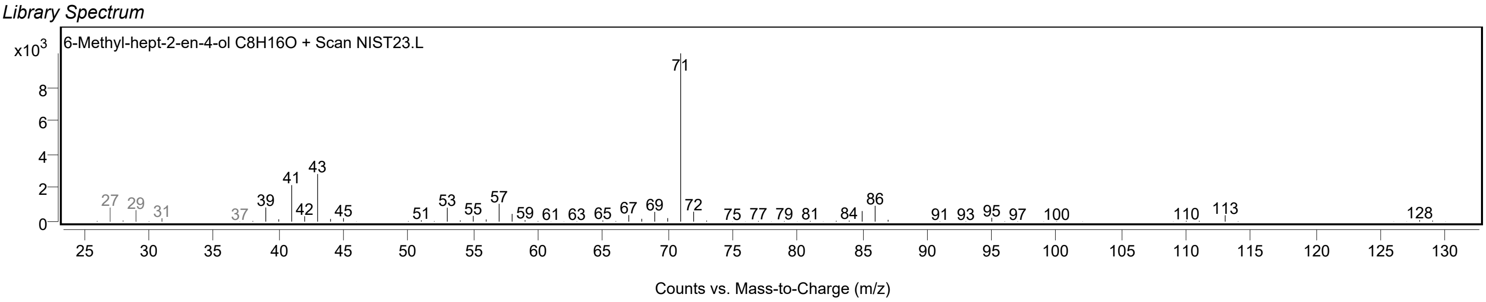
Observed Spectrum



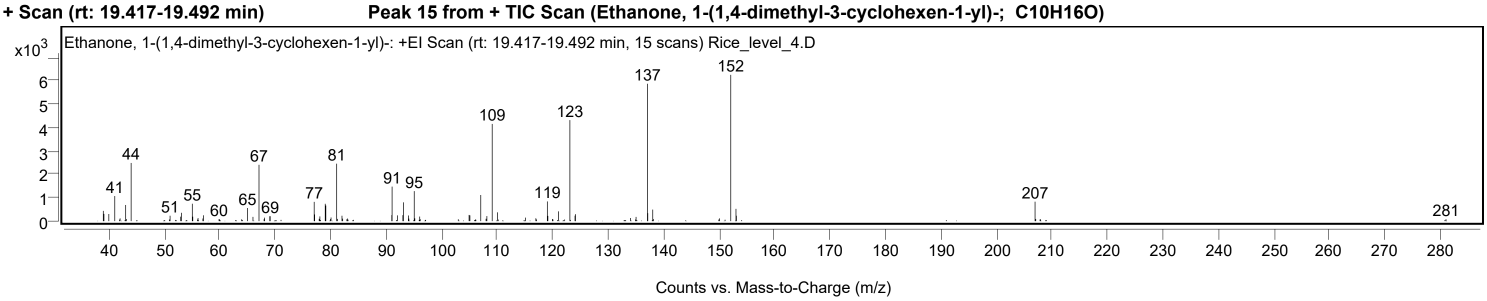
Mirror Plot



Library Spectrum



+ Scan (rt: 19.417-19.492 min) **Peak 15 from + TIC Scan (Ethanone, 1-(1,4-dimethyl-3-cyclohexen-1-yl)-; C10H16O)**



Spectrum Peaks

m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
38.9500		438	7.01					
39.0700		310	4.96					
39.9500		304	4.87					
41.0200		1072	17.15					
41.8800		67	1.07					
41.9800		118	1.88					
42.9900		690	11.04					
43.1000		91	1.46					
43.9700		2487	39.78					
50.8600		69	1.10					
50.9900		232	3.71					
52.9100		192	3.07					
53.0300		356	5.70					
55.0000		744	11.91					
55.0900		190	3.04					
55.9200		68	1.08					
56.0400		131	2.09					
56.9100		113	1.81					
57.0200		250	3.99					
59.8200		85	1.36					
59.9400		72	1.16					
63.8700		75	1.21					
64.9600		560	8.96					
65.9400		182	2.91					
67.0300	1	2408	38.52					
67.9400		84	1.34					
68.0400	1	148	2.37					
68.9700	1	209	3.34					
69.0500		210	3.35					
76.9800		824	13.19					
77.0600		284	4.55					
77.8800		100	1.59					
78.0100		203	3.25					
78.9700		742	11.86					
79.0800		667	10.68					
79.9000		87	1.40					
80.0200		155	2.48					
81.0400	1	2455	39.27					
81.1700		96	1.54					
81.9900	1	231	3.69					
82.1200		107	1.70					
82.8700		80	1.28					
82.9600	1	111	1.78					
83.0800		63	1.01					
90.9700		259	4.14					
91.0300		1475	23.60					
92.0300		246	3.94					
92.9700		255	4.08					
93.0800		804	12.86					
93.9600		242	3.87					
94.0600		108	1.73					
95.0200		1283	20.52					
95.1300		118	1.89					
96.0100		196	3.14					
96.1000		84	1.34					
102.9600		80	1.28					
104.9100		265	4.24					
105.0600		248	3.96					
106.0500		72	1.15					
107.0200	1	1117	17.87					
107.9700	1	100	1.60					
108.1200		217	3.47					
109.0800	1	4154	66.44					
110.0600	1	377	6.02					
110.1600		68	1.09					
115.0700		151	2.41					
116.9500		121	1.94					
117.0900		72	1.15					
119.0200	1	842	13.48					
119.1200		225	3.59					
119.9300		92	1.47					
120.0400	1	72	1.15					
121.0500	1	418	6.68					
122.1500		70	1.12					
123.0900	1	4318	69.06					
124.0100		234	3.74					
124.1000	1	296	4.73					
134.0000		134	2.15					
134.9100		69	1.10					
135.0200		181	2.90					
137.0800	1	5871	93.90					
137.1400		339	5.43					
138.0300	1	494	7.90					
138.1700		86	1.38					
149.9400		67	1.07					
150.0800		117	1.88					
152.1000	1	6252	100.00					
153.0400		525	8.39					
153.1200	1	231	3.69					
206.9700	1	832	13.31					
207.1000		93	1.49					
207.8700		72	1.15					
207.9600	1	78	1.25					

Spectrum Peaks

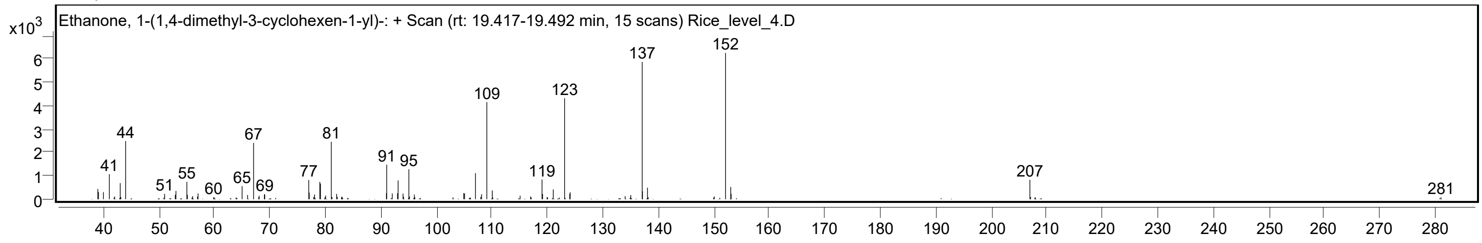
m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
281.0700		84	1.34					

Spectrum Identification Table

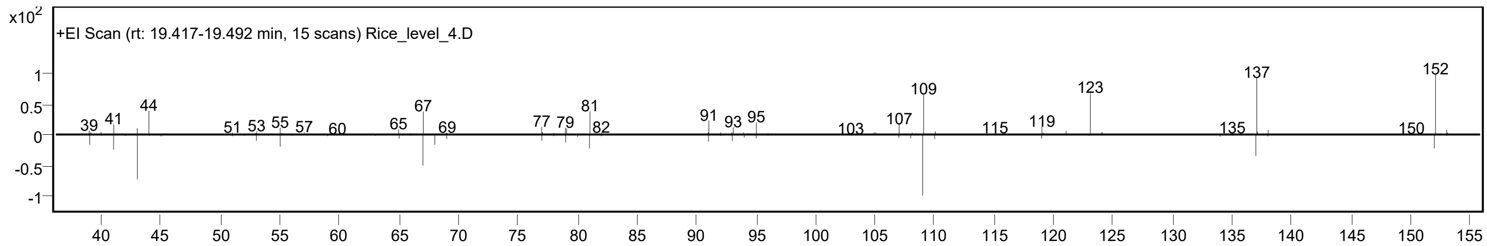
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	Ethanone, 1-(1,4-dimethyl-3-cyclohexen-1-yl)-	C10H16O				43219-68-7	76.38	76.38			NIST23.L
No	LibSearch	Ethanone, 1-(1,4-dimethyl-3-cyclohexen-1-yl)-	C10H16O				43219-68-7	74.98	74.98			NIST23.L
No	LibSearch	1-Cyclohexene-1-carboxaldehyde, 2,6,6-trimethyl-	C10H16O				432-25-7	71.03	71.03			NIST23.L
No	LibSearch	1-Cyclohexene-1-carboxaldehyde, 2,6,6-trimethyl-	C10H16O				432-25-7	66.13	66.13			NIST23.L
No	LibSearch	1-Cyclohexene-1-carboxaldehyde, 2,6,6-trimethyl-	C10H16O				432-25-7	65.65	65.65			NIST23.L
No	LibSearch	1-Cyclohexene-1-carboxaldehyde, 2,6,6-trimethyl-	C10H16O				432-25-7	65.03	65.03			NIST23.L
No	LibSearch	3-Cyclohexene-1-carboxaldehyde, 1,3,4-trimethyl-	C10H16O				40702-26-9	64.08	64.08			NIST23.L
No	LibSearch	4-Hydroxy-2,4,5-trimethyl-2,5-cyclohexadien-1-one	C9H12O2				14353-72-1	61.61	61.61			NIST23.L
No	LibSearch	3-Isopropylidene-5-methyl-hex-4-en-2-one	C10H16O				64149-32-2	59.83	59.83			NIST23.L
No	LibSearch	1,4,4-Trimethylcyclohex-2-enecarboxylic acid, methyl ester	C11H18O2				1000187-21-8	59.73	59.73			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	Ethanone, 1-(1,4-dimethyl-3-cyclohexen-1-yl)-	C10H16O		43219-68-7	19.467		76.38	76.38			NIST23.L

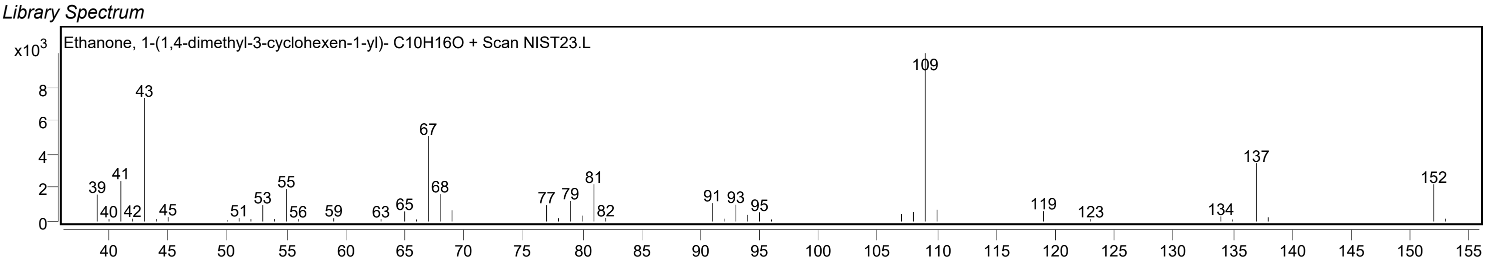
Observed Spectrum



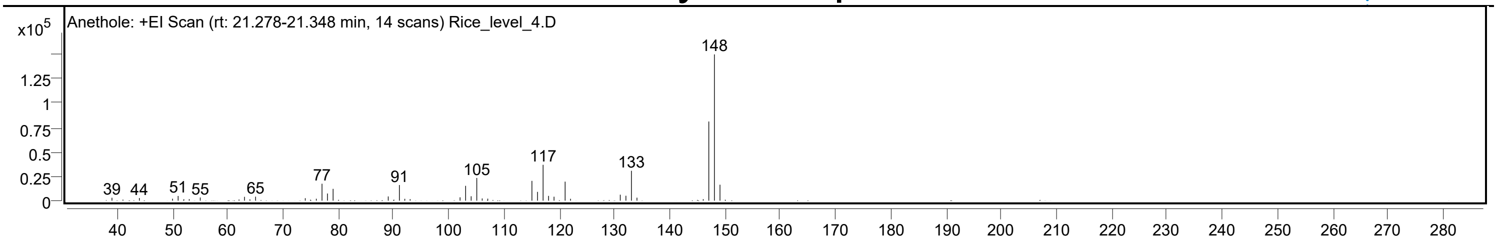
Mirror Plot



Library Spectrum



+ Scan (rt: 21.278-21.348 min) **Peak 16 from + TIC Scan (Anethole; C10H12O)**



Spectrum Peaks

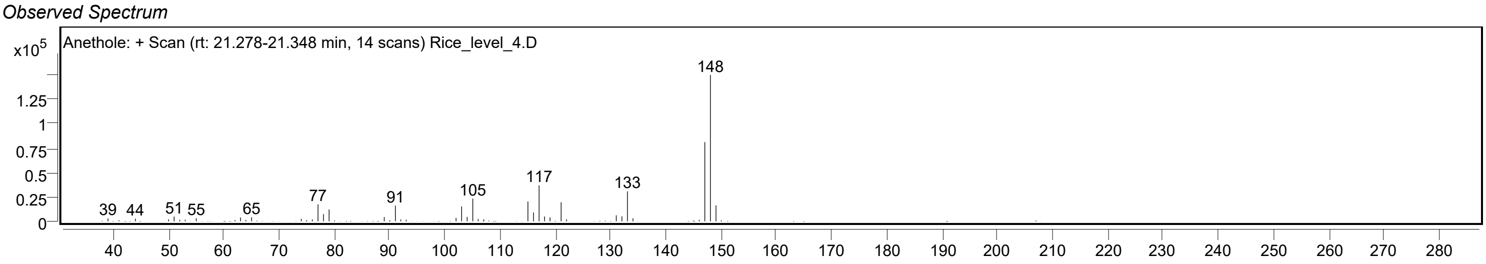
m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
39.0200		2993	2.00					
43.9800		2736	1.83					
49.9900		2019	1.35					
51.0000		4838	3.23					
52.0300		1507	1.01					
53.0100		1623	1.08					
55.0000		3246	2.17					
63.0100		3954	2.64					
65.0200		4091	2.73					
74.0100		2566	1.71					
76.0100		1990	1.33					
77.0400		17371	11.60					
78.0300		7342	4.90					
79.0500		12112	8.09					
89.0300		4349	2.90					
91.0500		16023	10.70					
92.0300		1913	1.28					
93.0000		1576	1.05					
102.0300		3476	2.32					
103.0400		15196	10.15					
104.0400		4527	3.02					
105.0600	1	23251	15.52					
106.0400	1	2300	1.54					
107.0500	1	2068	1.38					
115.0400		20281	13.54					
116.0500		9025	6.03					
117.0600	1	36759	24.54					
118.0500	1	4920	3.28					
119.0400	1	4063	2.71					
121.0500	1	19483	13.01					
122.0300	1	1935	1.29					
131.0300		6146	4.10					
132.0500		5060	3.38					
133.0500	1	30596	20.43					
134.0500	1	3095	2.07					
146.0500		1545	1.03					
147.0600		81062	54.12					
148.0700	1	149775	100.00					
149.0800	1	16360	10.92					

Spectrum Identification Table

Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	Anethole	C10H12O				104-46-1	74.22	74.22			NIST23.L
No	LibSearch	Anethole	C10H12O				104-46-1	73.99	73.99			NIST23.L
No	LibSearch	Anethole	C10H12O				104-46-1	73.99	73.99			NIST23.L
No	LibSearch	Anethole	C10H12O				104-46-1	73.97	73.97			NIST23.L
No	LibSearch	Benzene, pentamethyl-	C11H16				700-12-9	73.94	73.94			NIST23.L
No	LibSearch	Anethole, trans-	C10H12O				4180-23-8	73.68	73.68			NIST23.L
No	LibSearch	Benzene, pentamethyl-	C11H16				700-12-9	73.66	73.66			NIST23.L
No	LibSearch	Benzene, pentamethyl-	C11H16				700-12-9	73.60	73.60			NIST23.L
No	LibSearch	Estragole	C10H12O				140-67-0	72.74	72.74			NIST23.L
No	LibSearch	Anethole, trans-	C10H12O				4180-23-8	72.73	72.73			NIST23.L

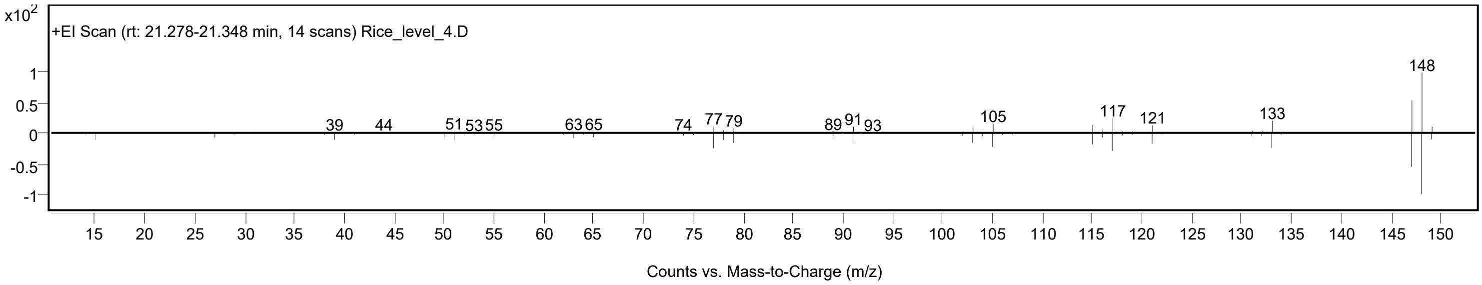
Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	Anethole	C10H12O		104-46-1	21.467		74.22	74.22			NIST23.L

Observed Spectrum

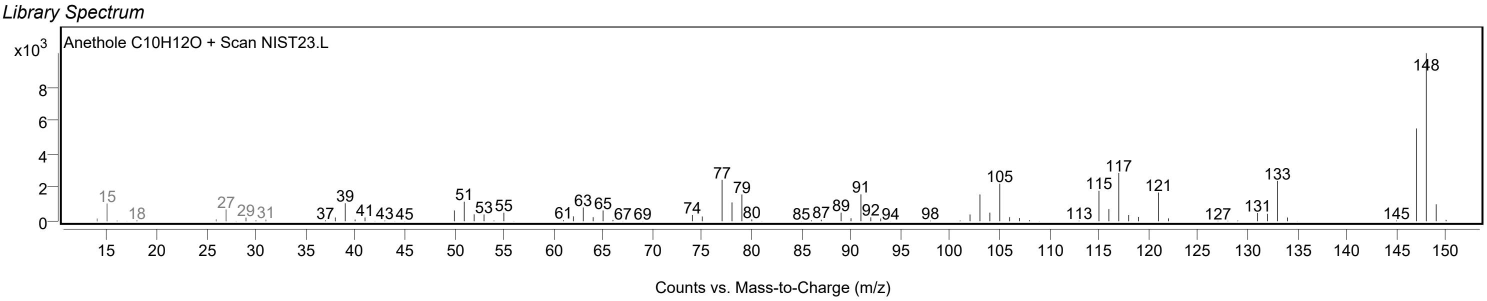


Analysis Report

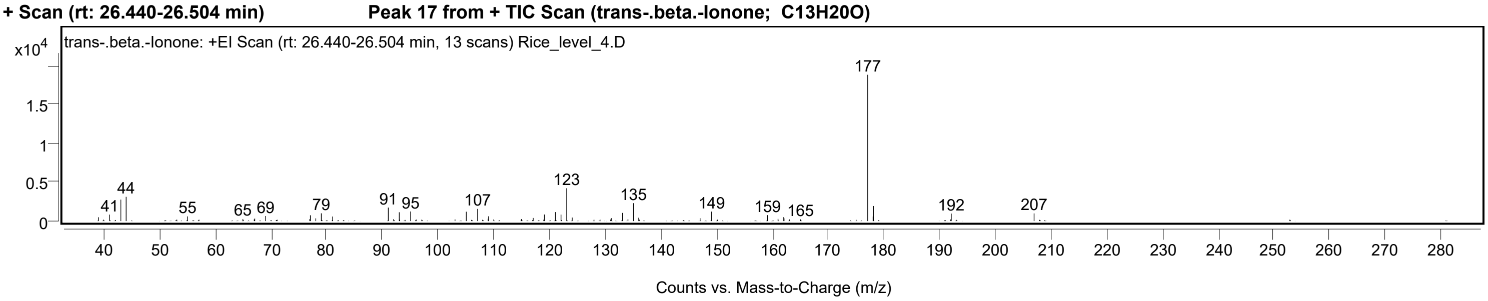
Mirror Plot



Library Spectrum



+ Scan (rt: 26.440-26.504 min) **Peak 17 from + TIC Scan (trans-.beta.-Ionone; C13H20O)**



Spectrum Peaks

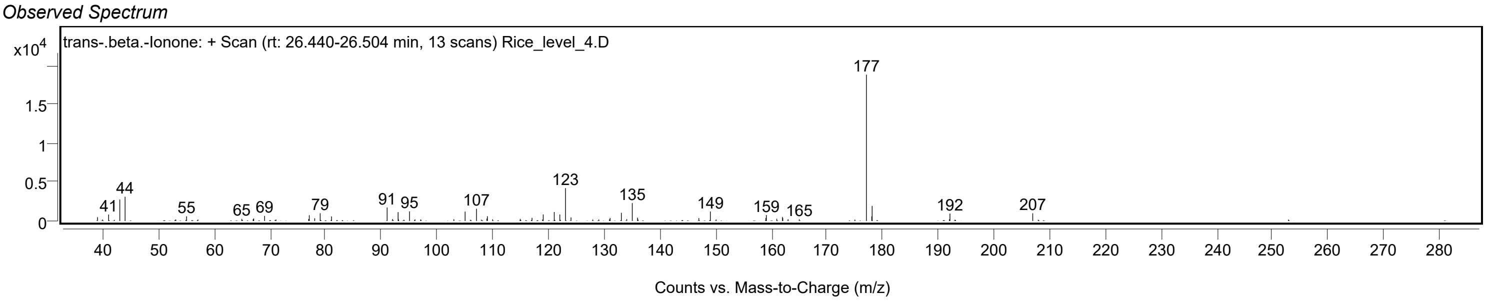
m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
39.0200		493	2.61					
41.0100		832	4.41					
43.0200		2769	14.65					
43.9700		3129	16.56					
53.0700		194	1.03					
55.0100		599	3.17					
64.9300		273	1.45					
66.9600		219	1.16					
67.0700		332	1.76					
69.0200		658	3.48					
76.9400		254	1.34					
77.0500		748	3.96					
78.0200		351	1.86					
79.0100		1000	5.29					
81.0500		587	3.11					
91.0300		1740	9.21					
92.0000		244	1.29					
93.0000		1139	6.03					
95.0500		1237	6.55					
96.0400		195	1.03					
103.0100		220	1.16					
105.0300		1209	6.40					
107.0700		1573	8.32					
108.9500		312	1.65					
109.0400		618	3.27					
109.9700		212	1.12					
114.9300		270	1.43					
117.0500		406	2.15					
119.0600		842	4.46					
121.0400		1163	6.16					
122.0400		832	4.40					
123.0600	1	4240	22.44					
124.0500	1	454	2.40					
130.9500		196	1.04					
131.0700		387	2.05					
133.0900		1057	5.59					
134.0600		239	1.26					
135.0500		2314	12.25					
135.9900		429	2.27					
147.0100		372	1.97					
149.0800		1222	6.47					
159.0400		401	2.12					
159.1300		768	4.06					
161.0000		285	1.51					
162.0200		510	2.70					
162.0900		350	1.85					
163.0300		216	1.14					
165.0400		218	1.15					
177.1000	1	18896	100.00					
178.0600		592	3.13					
178.1300	1	1949	10.31					
192.0000		210	1.11					
192.1000		964	5.10					
206.9700		990	5.24					

Spectrum Identification Table

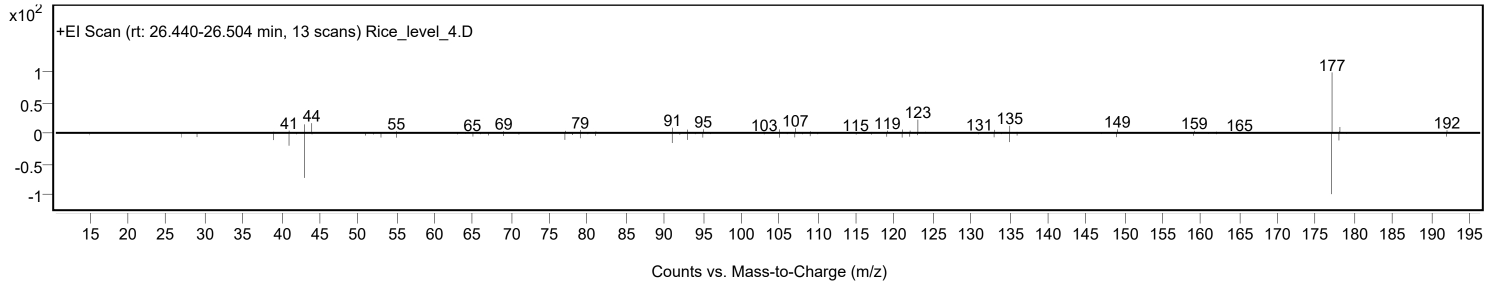
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	trans-.beta.-Ionone	C13H20O				79-77-6	66.85	66.85			NIST23.L
No	LibSearch	trans-.beta.-Ionone	C13H20O				79-77-6	65.97	65.97			NIST23.L
No	LibSearch	trans-.beta.-Ionone	C13H20O				79-77-6	65.56	65.56			NIST23.L
No	LibSearch	trans-.beta.-Ionone	C13H20O				79-77-6	64.01	64.01			NIST23.L
No	LibSearch	trans-.beta.-Ionone	C13H20O				79-77-6	63.60	63.60			NIST23.L
No	LibSearch	3-Buten-2-one, 4-(2,6,6-trimethyl-1-cyclohexen-1-yl)-	C13H20O				14901-07-6	63.51	63.51			NIST23.L
No	LibSearch	trans-.beta.-Ionone	C13H20O				79-77-6	62.27	62.27			NIST23.L
No	LibSearch	3-Buten-2-one, 4-(2,6,6-trimethyl-1-cyclohexen-1-yl)-	C13H20O				14901-07-6	62.23	62.23			NIST23.L
No	LibSearch	3-Buten-2-one, 4-(2,6,6-trimethyl-1-cyclohexen-1-yl)-	C13H20O				14901-07-6	61.33	61.33			NIST23.L
No	LibSearch	trans-.beta.-Ionone	C13H20O				79-77-6	60.71	60.71			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	trans-.beta.-Ionone	C13H20O		79-77-6	24.467		66.85	66.85			NIST23.L

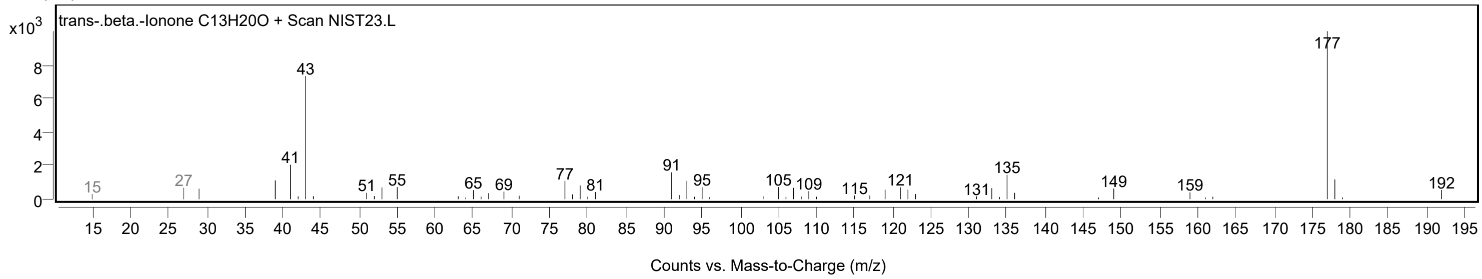
Observed Spectrum



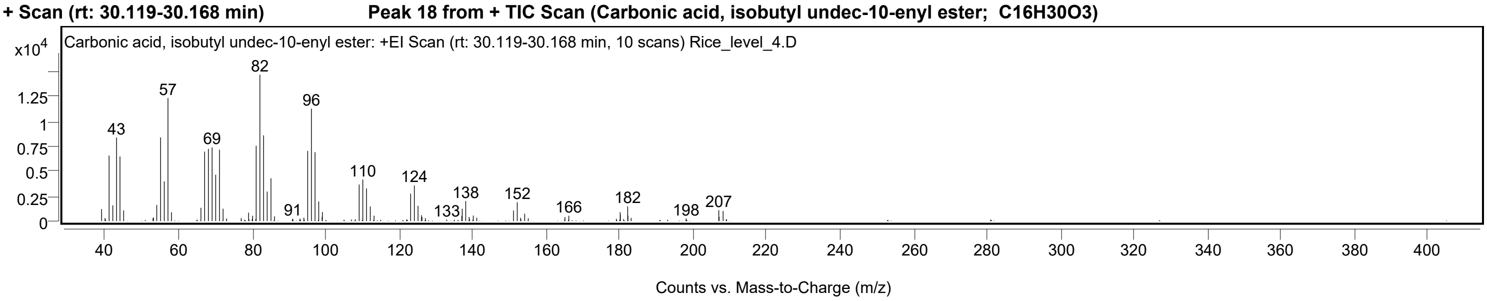
Mirror Plot



Library Spectrum



+ Scan (rt: 30.119-30.168 min) **Peak 18 from + TIC Scan (Carbonic acid, isobutyl undec-10-enyl ester; C16H30O3)**



Spectrum Peaks

m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
38.9900		1206	8.20					
39.9200		304	2.07					
40.0700		160	1.09					
41.0400		6586	44.76					
42.0500		1585	10.77					
43.0700		8400	57.09					
44.0000		6497	44.16					
45.0000		1081	7.34					
52.9800		278	1.89					
53.1100		372	2.53					
54.0100		1619	11.00					
55.0500		8413	57.18					
56.0500		3997	27.16					
57.0600	1	12370	84.07					
58.0200	1	883	6.00					
66.0300		1333	9.06					
67.0400		6983	47.46					
68.0600		7257	49.32					
69.0700		7404	50.32					
70.0500		4656	31.65					
71.0800		7176	48.77					
72.0500		1246	8.47					
73.0100		254	1.73					
76.9900		301	2.05					
79.0000		844	5.74					
79.1000		172	1.17					
79.9700		195	1.32					
80.0900		533	3.62					
80.9500		186	1.27					
81.0600		7574	51.48					
82.0800		14713	100.00					
83.0800		8622	58.60					
84.0600		2967	20.17					
85.0900		4304	29.25					
86.0500		474	3.22					
90.9100		149	1.01					
91.0300		238	1.62					
92.9700		236	1.61					
94.0400		334	2.27					
95.0800		7052	47.93					
96.0900		11307	76.85					
97.0800		6931	47.11					
98.0900		1972	13.40					
99.0000		437	2.97					
99.0800		901	6.12					
107.0200		159	1.08					
108.0400		183	1.24					
109.0900		3676	24.98					
110.0800		4176	28.38					
111.0900		3283	22.31					
112.1100		1476	10.03					
113.1400		540	3.67					
122.0400		166	1.13					
123.1000		2764	18.79					
124.1000		3581	24.34					
125.1100		1535	10.43					
126.0400		587	3.99					
126.1900		400	2.72					
127.1200		298	2.03					
132.9300		153	1.04					
137.0300		515	3.50					
137.1300		1245	8.46					
138.0900		2016	13.70					
139.0200		410	2.79					
139.1200		190	1.29					
140.1400		556	3.78					
141.0900		347	2.36					
151.0900		1059	7.20					
152.1000	1	1897	12.89					
153.0200		328	2.23					
153.1200	1	164	1.11					
154.1300	1	741	5.03					
155.1000		285	1.94					
165.0900		399	2.71					
166.1400		536	3.64					
179.0700		219	1.49					
180.1100	1	895	6.08					
180.1700		652	4.43					
181.0500	1	194	1.32					
182.1600	1	1479	10.05					
182.2300		544	3.70					
183.1000	1	317	2.16					
198.0600		256	1.74					
206.9500		1100	7.48					
207.0300		430	2.93					
208.1300	1	1015	6.90					
208.9800	1	181	1.23					

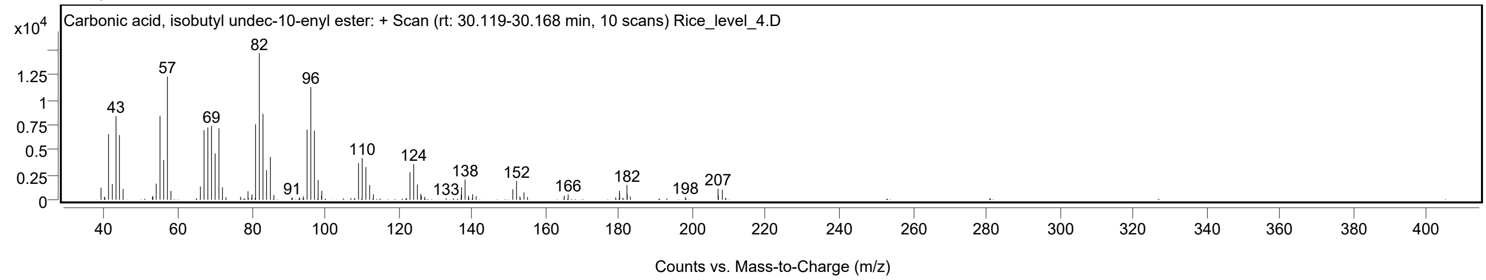
Analysis Report

Spectrum Identification Table

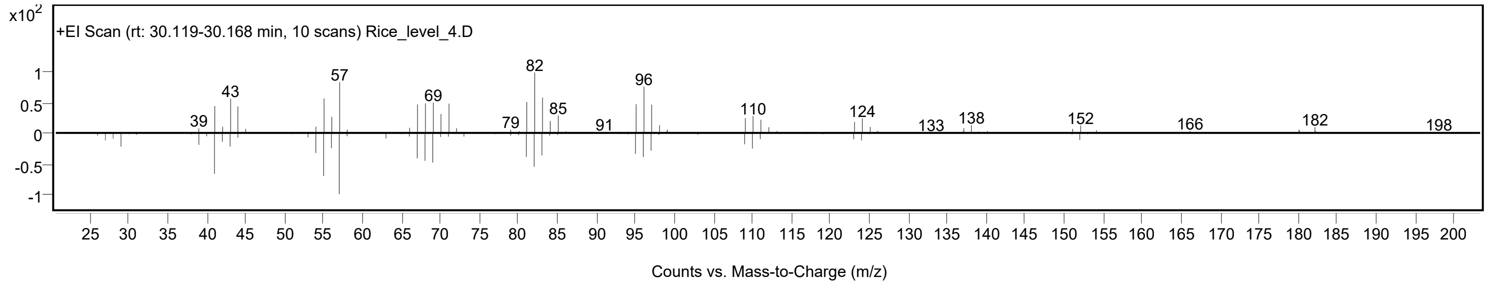
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	Carbonic acid, isobutyl undec-10-enyl ester	C16H30O3				1000314-60-8	79.72	79.72			NIST23.L
No	LibSearch	trans-11-Tetradecenyl acetate	C16H30O2				33189-72-9	79.26	79.26			NIST23.L
No	LibSearch	11-Tetradecen-1-ol, acetate, (Z)-	C16H30O2				20711-10-8	79.24	79.24			NIST23.L
No	LibSearch	11-Tetradecen-1-ol, acetate, (Z)-	C16H30O2				20711-10-8	78.68	78.68			NIST23.L
No	LibSearch	trans-11-Tetradecenyl acetate	C16H30O2				33189-72-9	77.66	77.66			NIST23.L
No	LibSearch	13-Tetradecen-1-ol acetate	C16H30O2				56221-91-1	77.61	77.61			NIST23.L
No	LibSearch	Pentadecanal-	C15H30O				2765-11-9	75.83	75.83			NIST23.L
No	LibSearch	11-Tetradecen-1-ol, acetate, (Z)-	C16H30O2				20711-10-8	75.81	75.81			NIST23.L
No	LibSearch	Cyclopentadecanol	C15H30O				4727-17-7	73.71	73.71			NIST23.L
No	LibSearch	Oxirane, tetradecyl-	C16H32O				7320-37-8	73.27	73.27			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	Carbonic acid, isobutyl undec-10-enyl ester	C16H30O3		1000314-60-8	30.167		79.72	79.72			NIST23.L

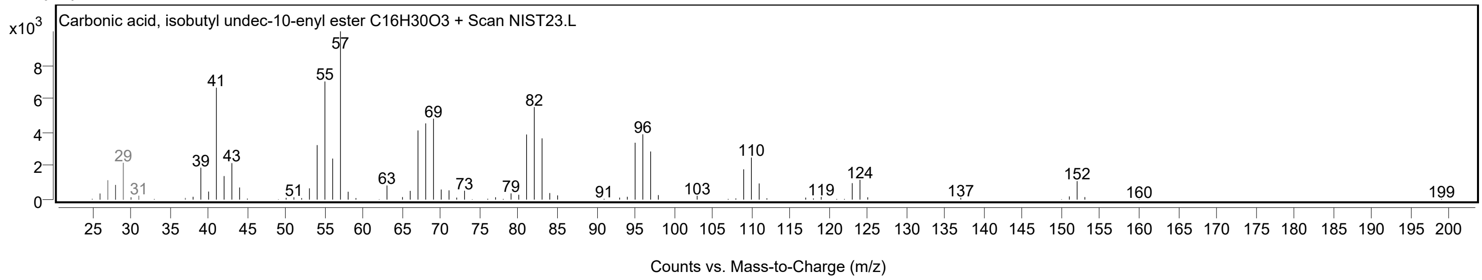
Observed Spectrum



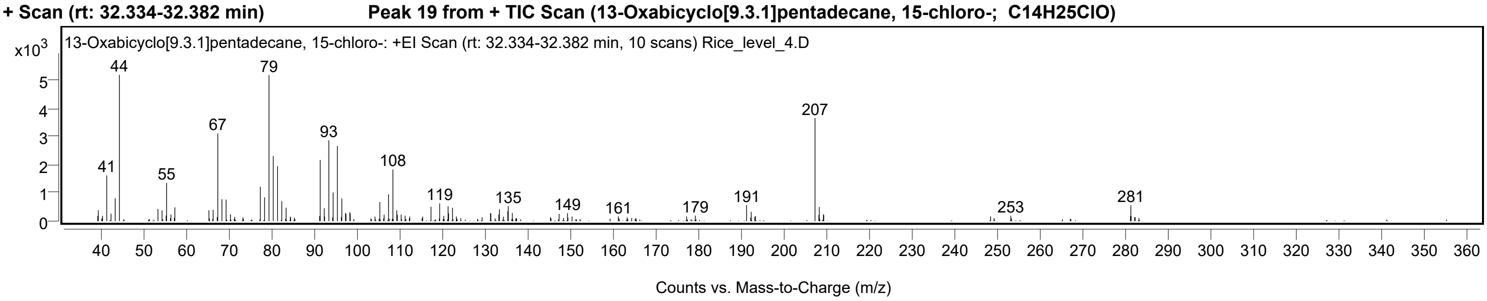
Mirror Plot



Library Spectrum



+ Scan (rt: 32.334-32.382 min) **Peak 19 from + TIC Scan (13-Oxabicyclo[9.3.1]pentadecane, 15-chloro-; C14H25ClO)**



Spectrum Peaks

m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
38.9200		183	3.54					
39.0500		387	7.47					
40.0500		185	3.58					
41.0300		1619	31.27					
42.0300		266	5.13					
43.0100		809	15.62					
43.9900		5177	100.00					
53.0300		431	8.32					
53.9900		378	7.30					
54.9500		195	3.76					
55.0600		1355	26.18					
56.0700		235	4.55					
56.8600		116	2.25					
56.9900		488	9.42					
64.9700		386	7.45					
65.9600		399	7.71					
66.9100		141	2.72					
67.0500		3103	59.94					
68.0200		776	14.99					
69.0000		764	14.75					
70.0300		243	4.70					
71.0100		152	2.94					
72.9400		135	2.60					
77.0100		1210	23.37					
78.0100		839	16.20					
79.0500		5163	99.72					
80.0400		2303	44.48					
81.0500		1944	37.55					
82.0400		707	13.66					
83.0600		475	9.17					
84.0400		156	3.01					
84.1300		118	2.27					
84.9800		123	2.37					
90.9200		181	3.49					
91.0500		2165	41.82					
91.9900		457	8.82					
92.0800		166	3.20					
93.0700	1	2858	55.21					
93.9600	1	130	2.52					
94.0800		1012	19.54					
95.0600	1	2657	51.31					
96.0000	1	142	2.73					
96.1100		802	15.49					
96.9700		288	5.57					
97.1000	1	255	4.92					
98.0000		316	6.10					
98.0900		260	5.02					
103.8800		157	3.04					
105.0200		677	13.08					
106.0300		227	4.39					
107.0600		951	18.37					
108.0800		1830	35.35					
108.9900		383	7.40					
109.1200		228	4.40					
110.0200		233	4.51					
110.9400		192	3.71					
112.0400		164	3.16					
115.0700		160	3.10					
117.0100		510	9.84					
119.0500		627	12.10					
119.9900		187	3.60					
121.0200		525	10.14					
121.1400		271	5.24					
122.0500		471	9.10					
122.9400		166	3.20					
128.9900		132	2.54					
130.9500		284	5.49					
131.0400		273	5.27					
132.9000		242	4.67					
133.0500		422	8.16					
133.9900		161	3.11					
135.0000		354	6.84					
135.1100		532	10.27					
136.0600		288	5.57					
145.0000		142	2.74					
147.0300		259	5.00					
149.0000		283	5.47					
149.1100		117	2.26					
150.0700		162	3.13					
160.9000		164	3.17					
162.9800		138	2.66					
176.9200		168	3.25					
178.9400		197	3.81					
190.9700	1	570	11.01					
191.9200	1	123	2.37					
192.0400		329	6.36					
192.8800		149	2.87					
193.0300	1	181	3.50					
207.0000	1	3649	70.47					
207.8900		220	4.24					
207.9900	1	500	9.65					
208.9600	1	243	4.70					
209.0500		208	4.02					

Spectrum Peaks

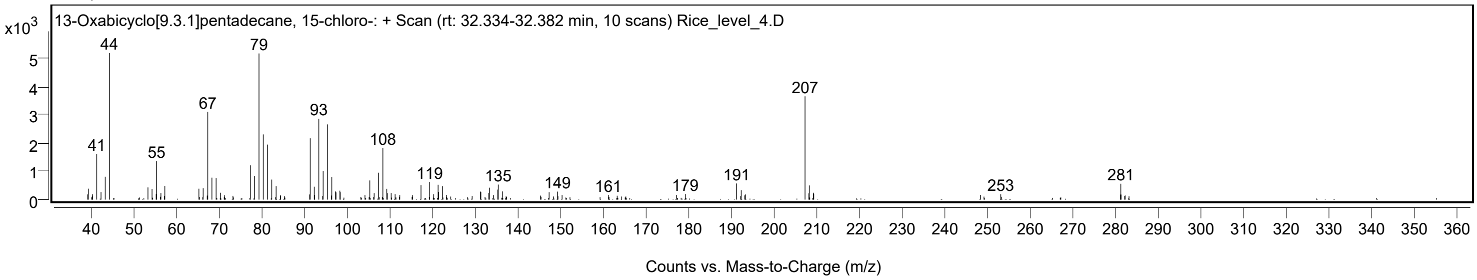
m/z	Z	Abund	Abund %	m/z (Calc)	Diff (ppm)	Ion Species	Formula	Ion Type
248.1500		169	3.27					
252.8700		188	3.63					
280.9100		178	3.44					
281.0000	1	561	10.84					
281.0700		179	3.45					
281.9000		130	2.52					
282.0200	1	145	2.80					

Spectrum Identification Table

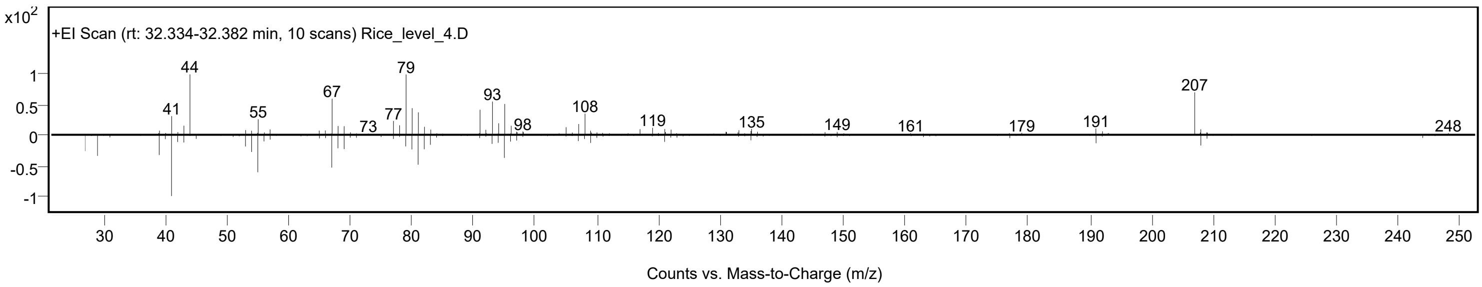
Best	ID Source	Name	Formula	Species	m/z	Diff (ppm)	CAS	Score	Score (Lib)	Score (DB)	Score (MFG)	Lib/DB
Yes	LibSearch	13-Oxabicyclo[9.3.1]pentadecane, 15-chloro-	C14H25ClO				1000151-77-6	65.46	65.46			NIST23.L
No	LibSearch	9,12,15-Octadecatrien-1-ol, (Z,Z,Z)-	C18H32O				506-44-5	54.79	54.79			NIST23.L
No	LibSearch	L-(+)-Camphorsultam	C10H17NO2S				108448-77-7	53.81	53.81			NIST23.L
No	LibSearch	Caryophyllene,14-hydroxy-4,5-dihydro-	C15H26O				289621-83-6	53.27	53.27			NIST23.L
No	LibSearch	1,8,11,14-Heptadecatetraene, (Z,Z,Z)-	C17H28				10482-53-8	52.73	52.73			NIST23.L
No	LibSearch	cis,cis,cis-7,10,13-Hexadecatrienal	C16H26O				56797-43-4	51.91	51.91			NIST23.L
No	LibSearch	1H-3a,7-Methanoazulene, octahydro-1,4,9,9-tetramethyl-	C15H26				25491-20-7	51.76	51.76			NIST23.L
No	LibSearch	9,12,15-Octadecatrienoic acid, (Z,Z,Z)-	C18H30O2				463-40-1	51.38	51.38			NIST23.L
No	LibSearch	3-Heptadecen-5-yne, (Z)-	C17H30				74744-55-1	51.36	51.36			NIST23.L
No	LibSearch	9,12,15-Octadecatrienoic acid, methyl ester, (Z,Z,Z)-	C19H32O2				301-00-8	51.31	51.31			NIST23.L

Best	Name	Formula	m/z (prec)	CAS	RT (DB)	RT Diff	Score	Score (Lib)	Score (Fwd)	Score (Rev)	Lib/DB
Yes	13-Oxabicyclo[9.3.1]pentadecane, 15-chloro-	C14H25ClO		1000151-77-6	32.367		65.46	65.46			NIST23.L

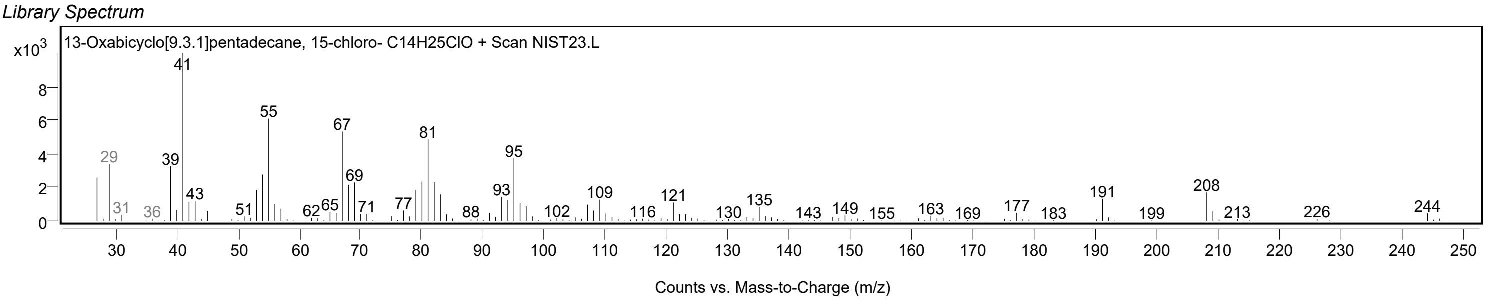
Observed Spectrum



Mirror Plot



Library Spectrum



MassHunter Qual 12.0
(End of Report)